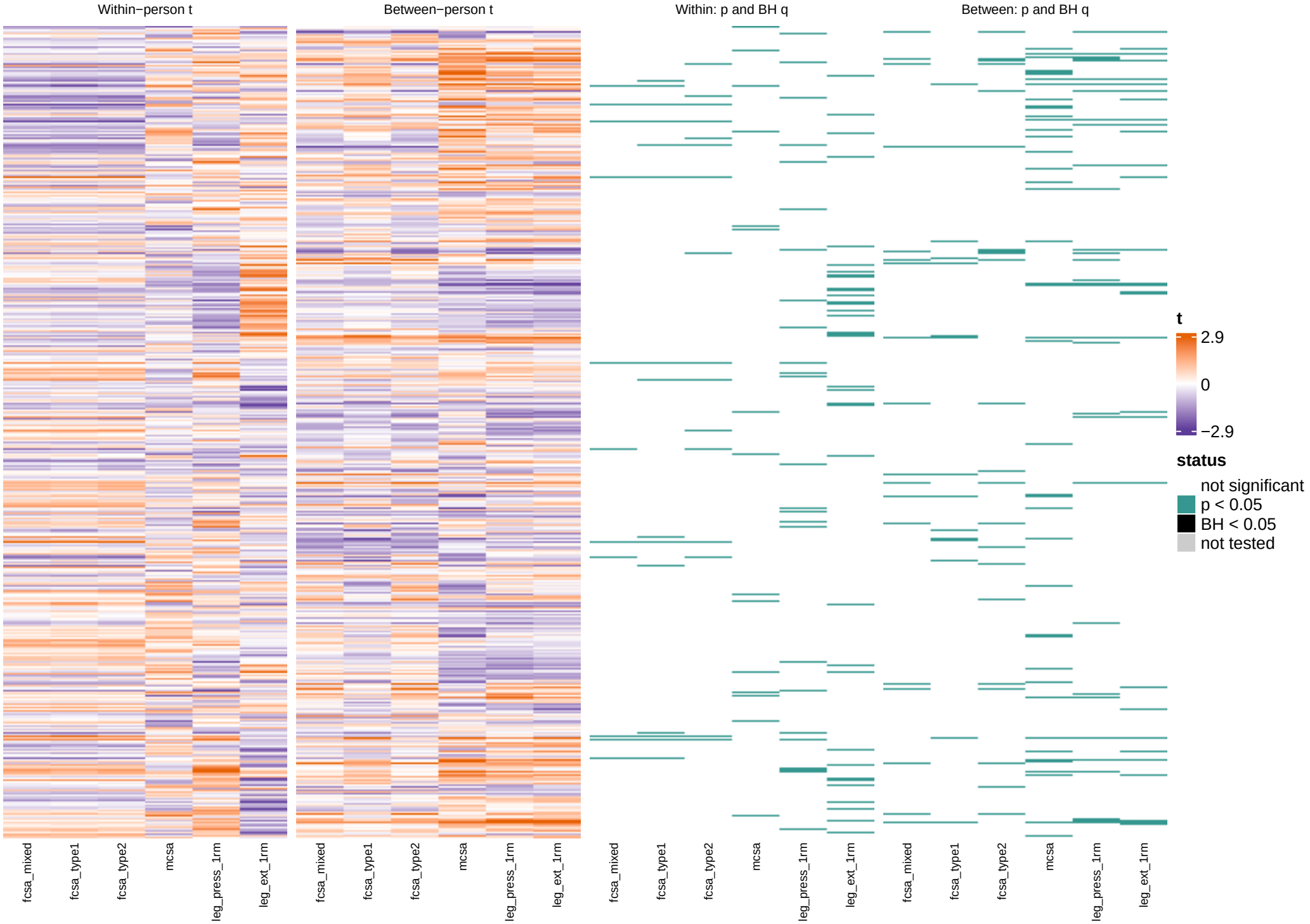
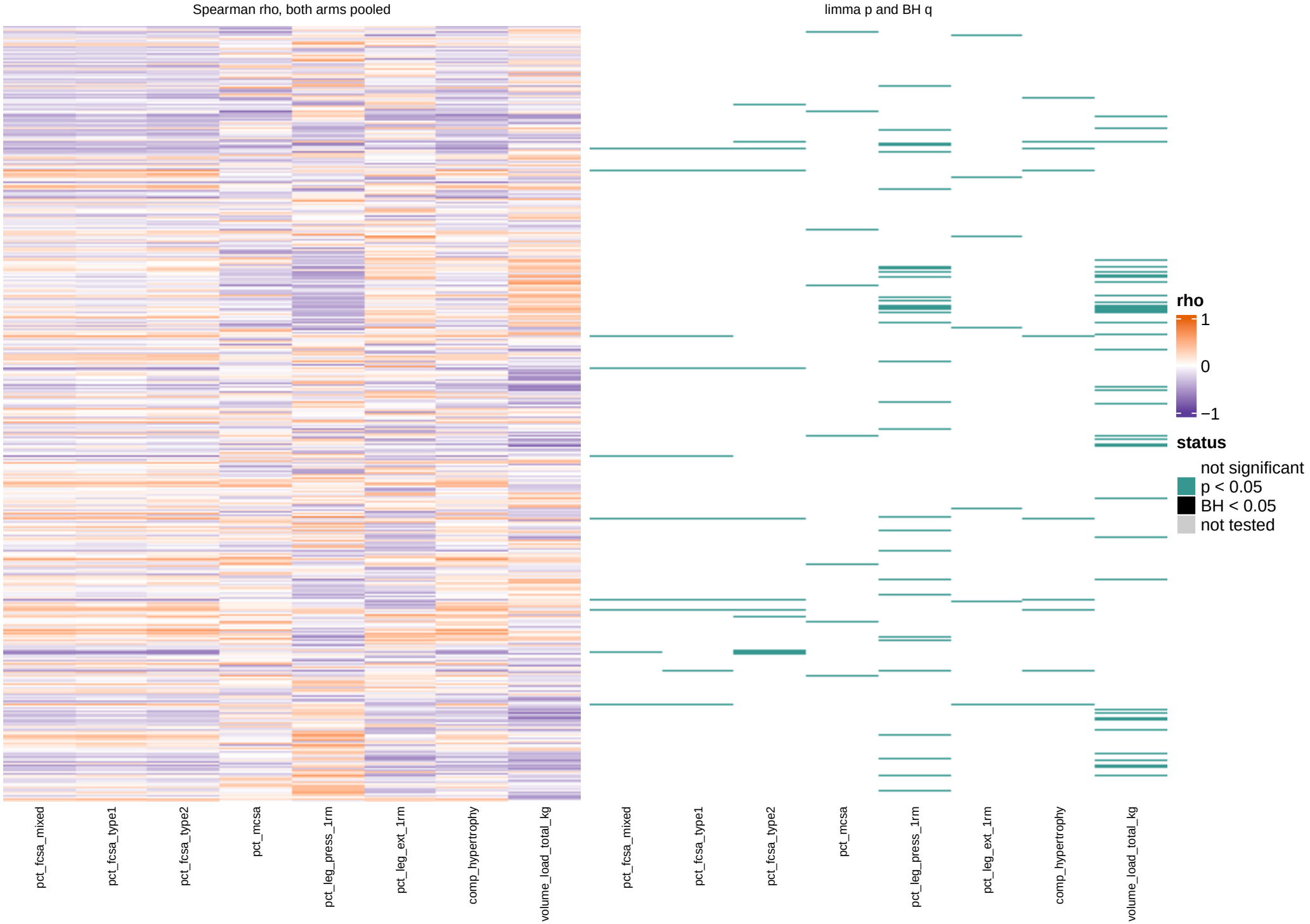
Sample model overview: proteins



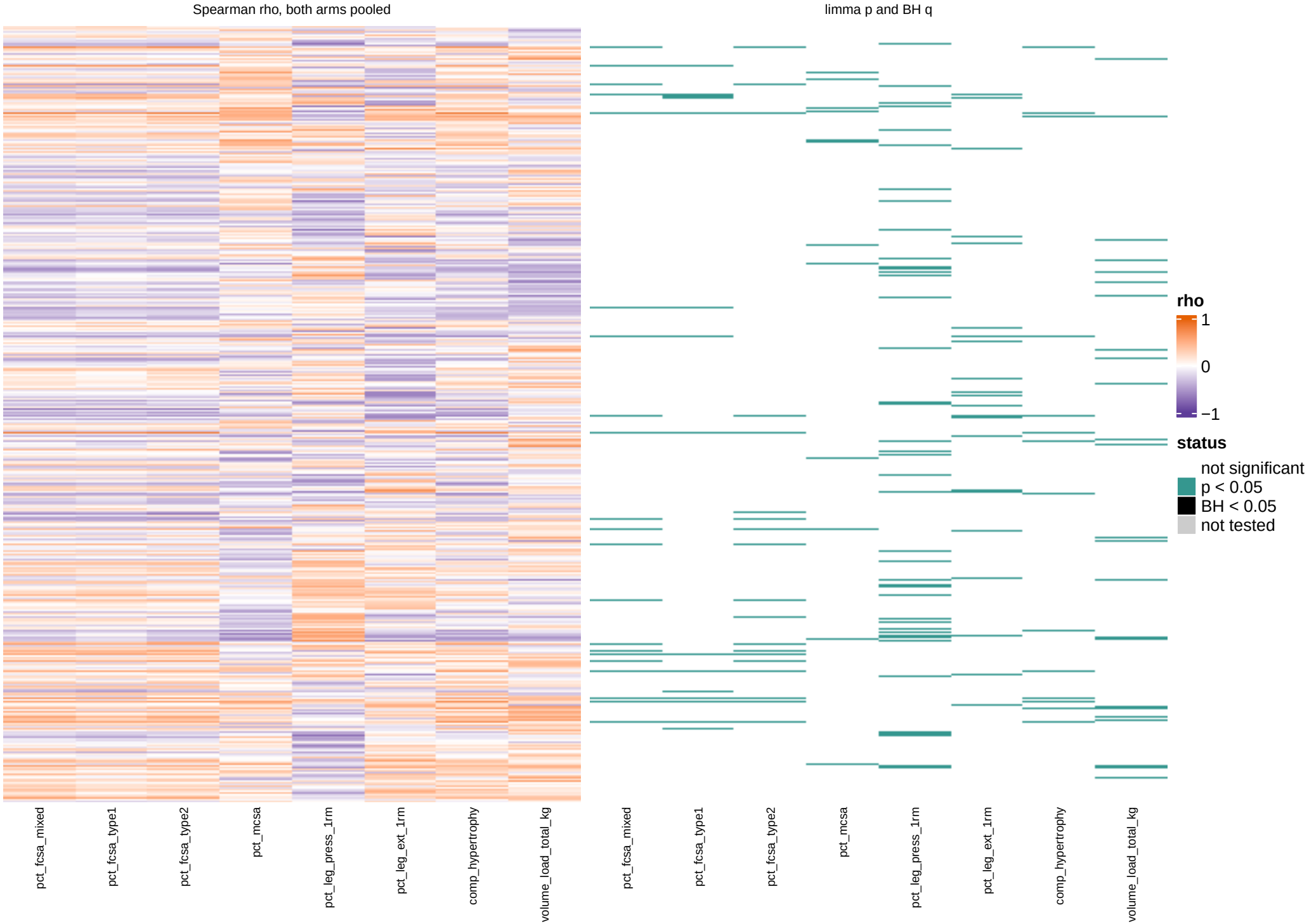
All 1,900 proteins, one row each. Moderated t of the within and between terms, abundance ~ timepoint + between + within, subject blocked. Colours stop at the 99th percentile of |t|, 2.9.
Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, model.

Change score overview, training window: proteins



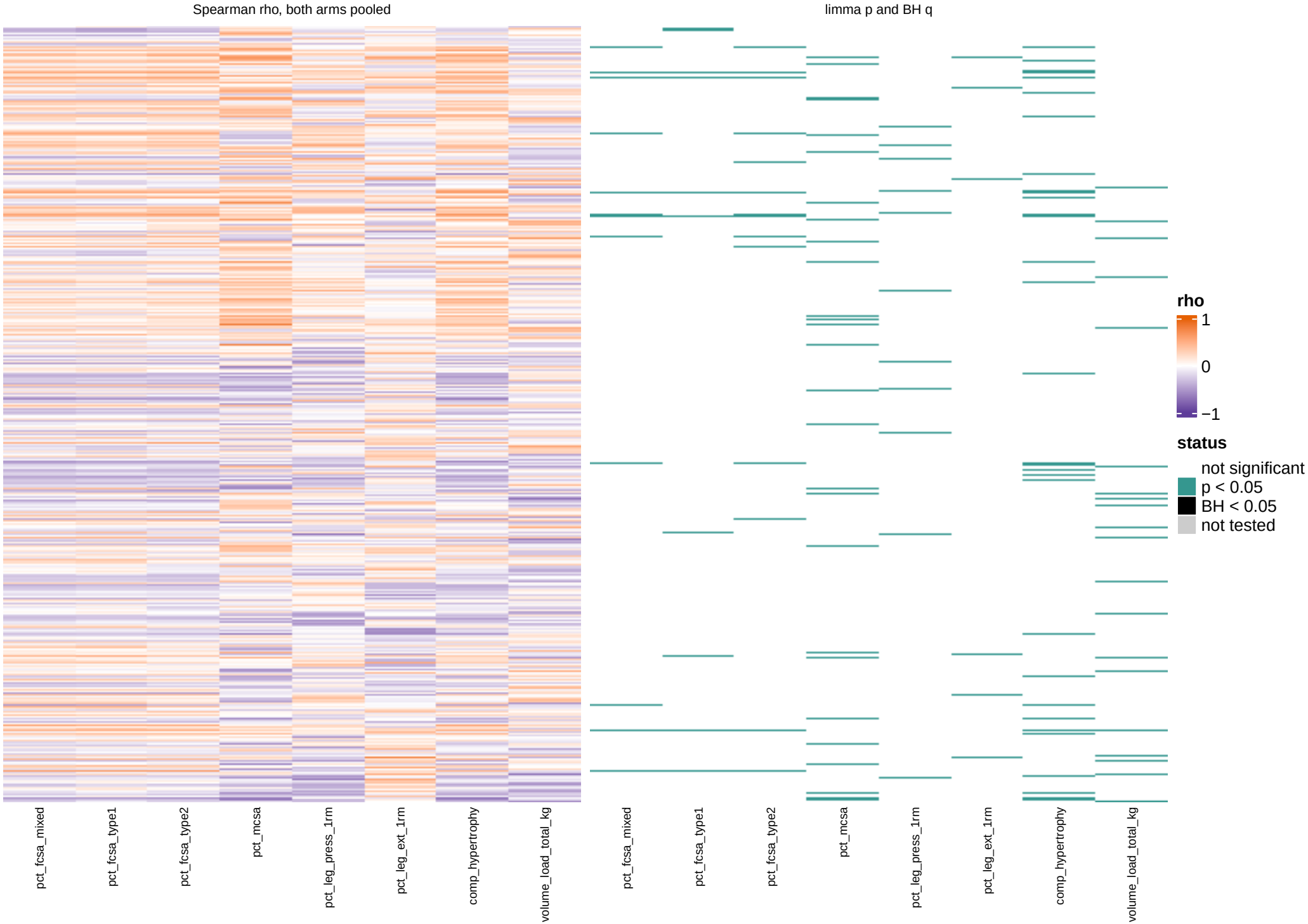
All 1,900 proteins, one row each. Spearman rho between the log2 abundance (change, T1 to T2) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

Change score overview, baseline window: proteins



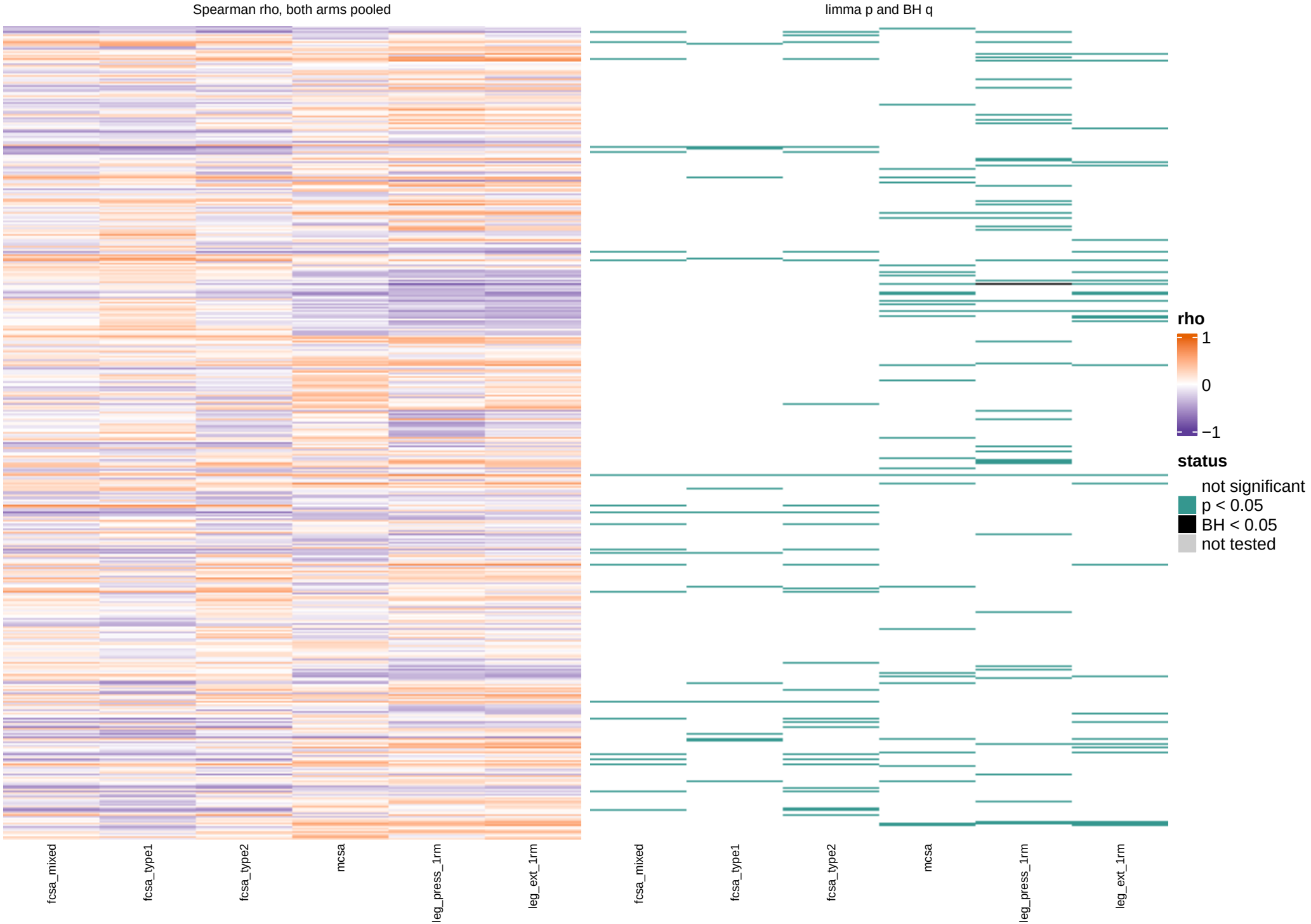
All 1,900 proteins, one row each. Spearman rho between the log2 abundance (T1 level) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

Change score overview, acute window: proteins



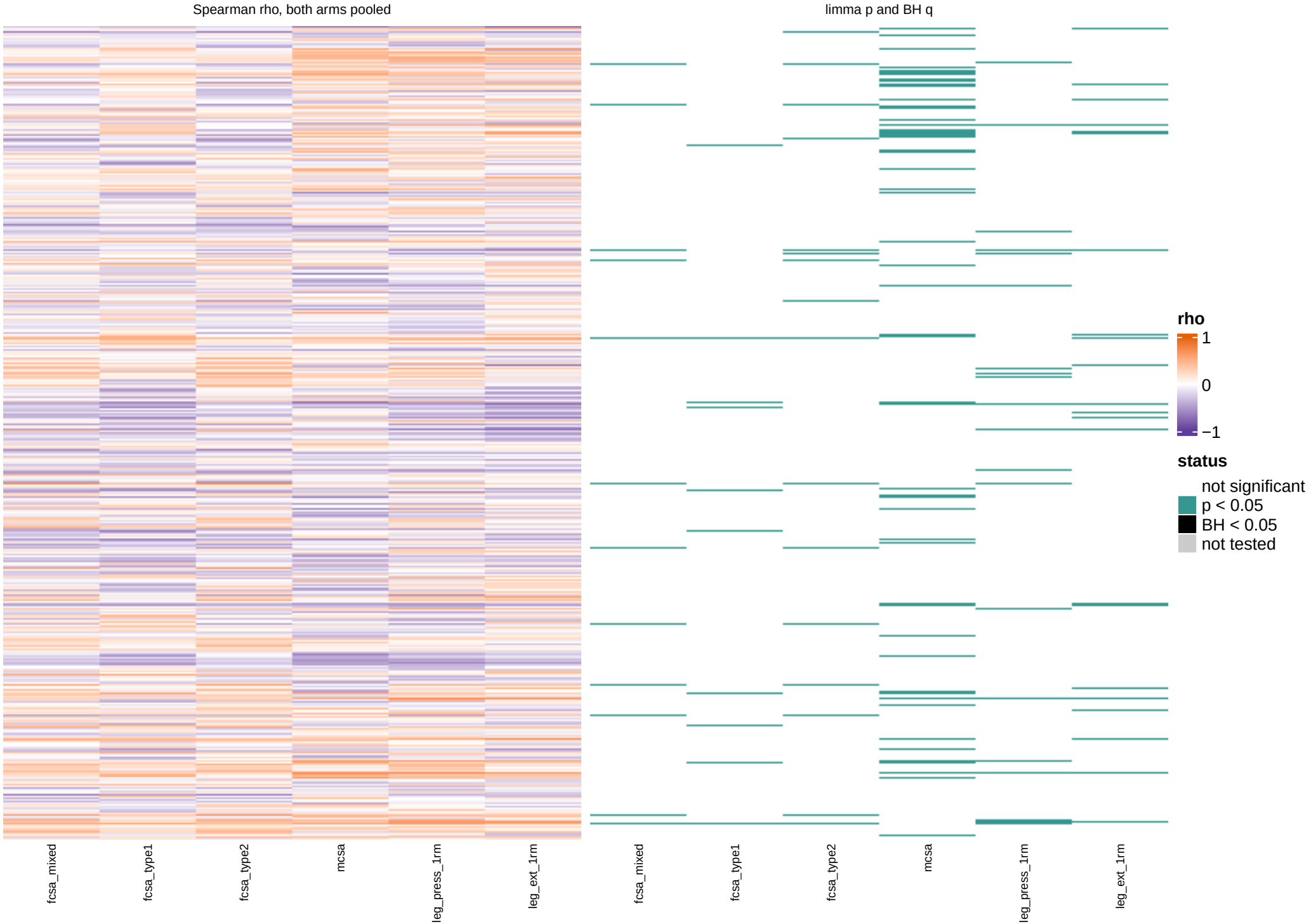
All 1,900 proteins, one row each. Spearman rho between the log2 abundance (change, T2 to T3) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

Change score overview, baseline_level window: proteins



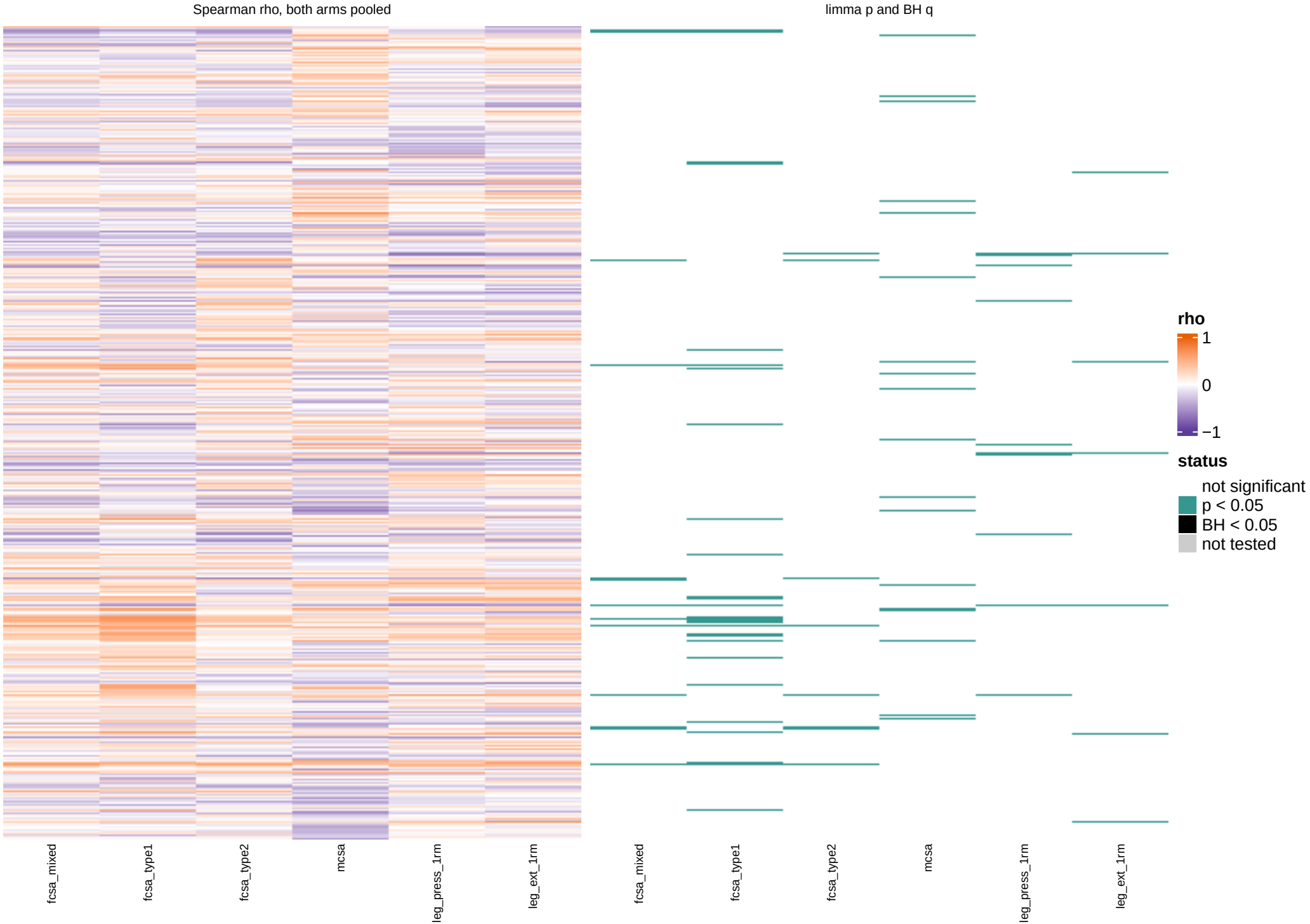
All 1,900 proteins, one row each. Spearman rho between the log2 abundance (T1 level) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

Change score overview, post_level window: proteins



All 1,900 proteins, one row each. Spearman rho between the log2 abundance (T2 level) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

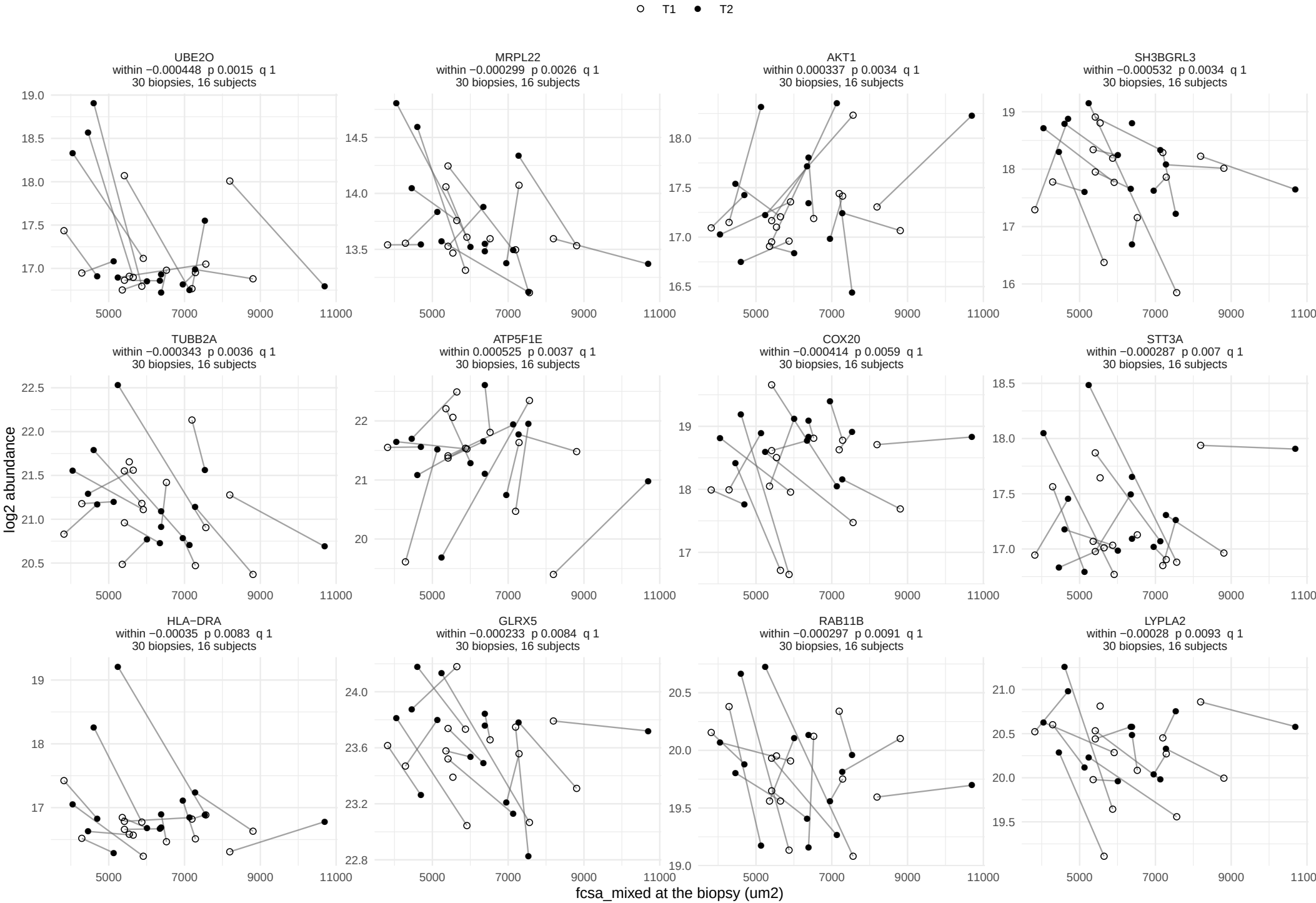
Change score overview, post_bout_level window: proteins



All 1,900 proteins, one row each. Spearman rho between the log2 abundance (T3 level) and each outcome; status from limma. Rows are ordered by average−linkage clustering (1 − Pearson r) of limpa's matrix, for display only. Grey: not tested. Table: c_data/01_proteins.xlsx, change_score.

fcsa_mixed, within−person term

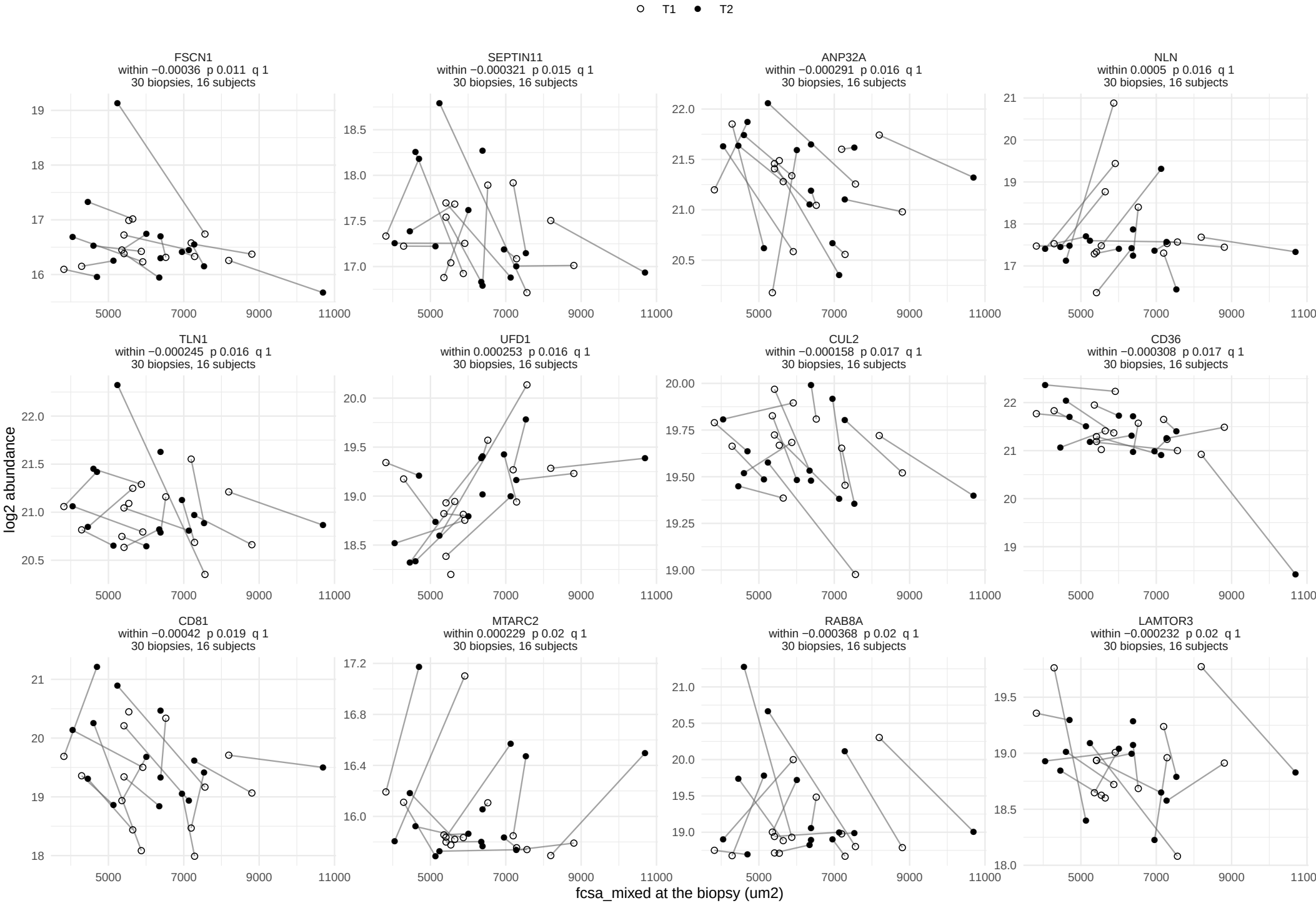
62 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_mixed, within−person term

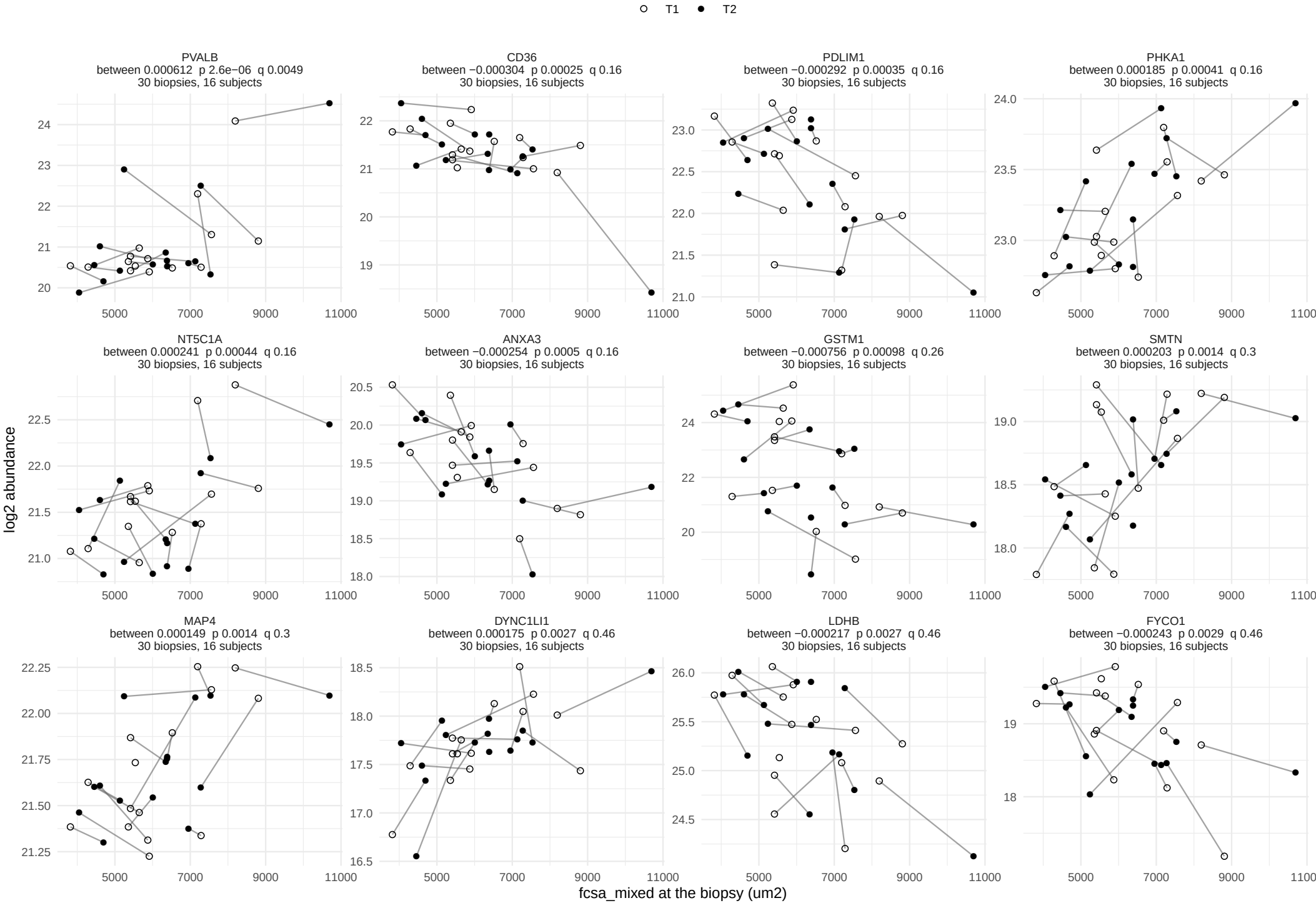
62 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_mixed, between−person term

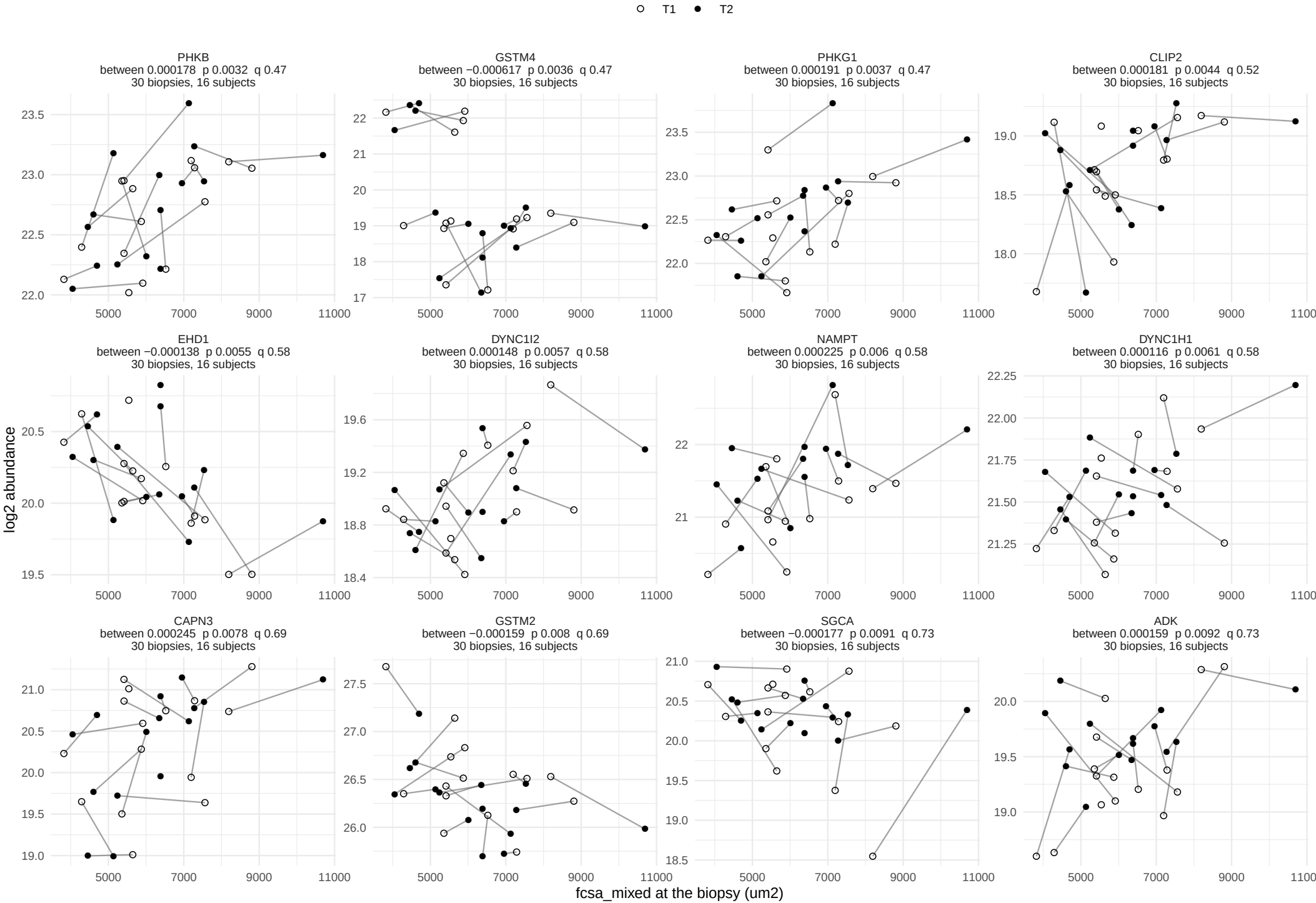
74 of 1900 proteins at p < 0.05, 1 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_mixed, between−person term

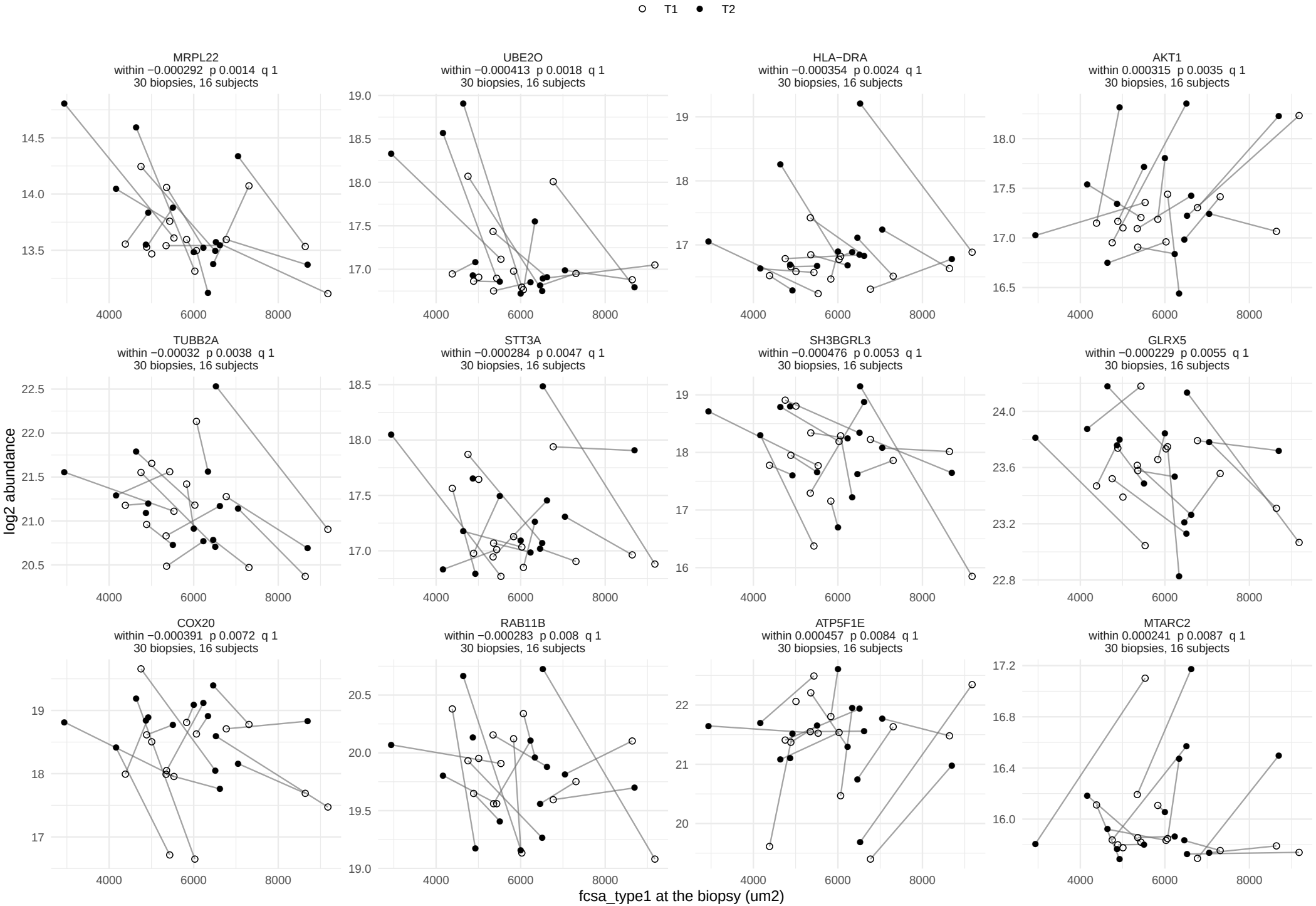
74 of 1900 proteins at p < 0.05, 1 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type1, within−person term

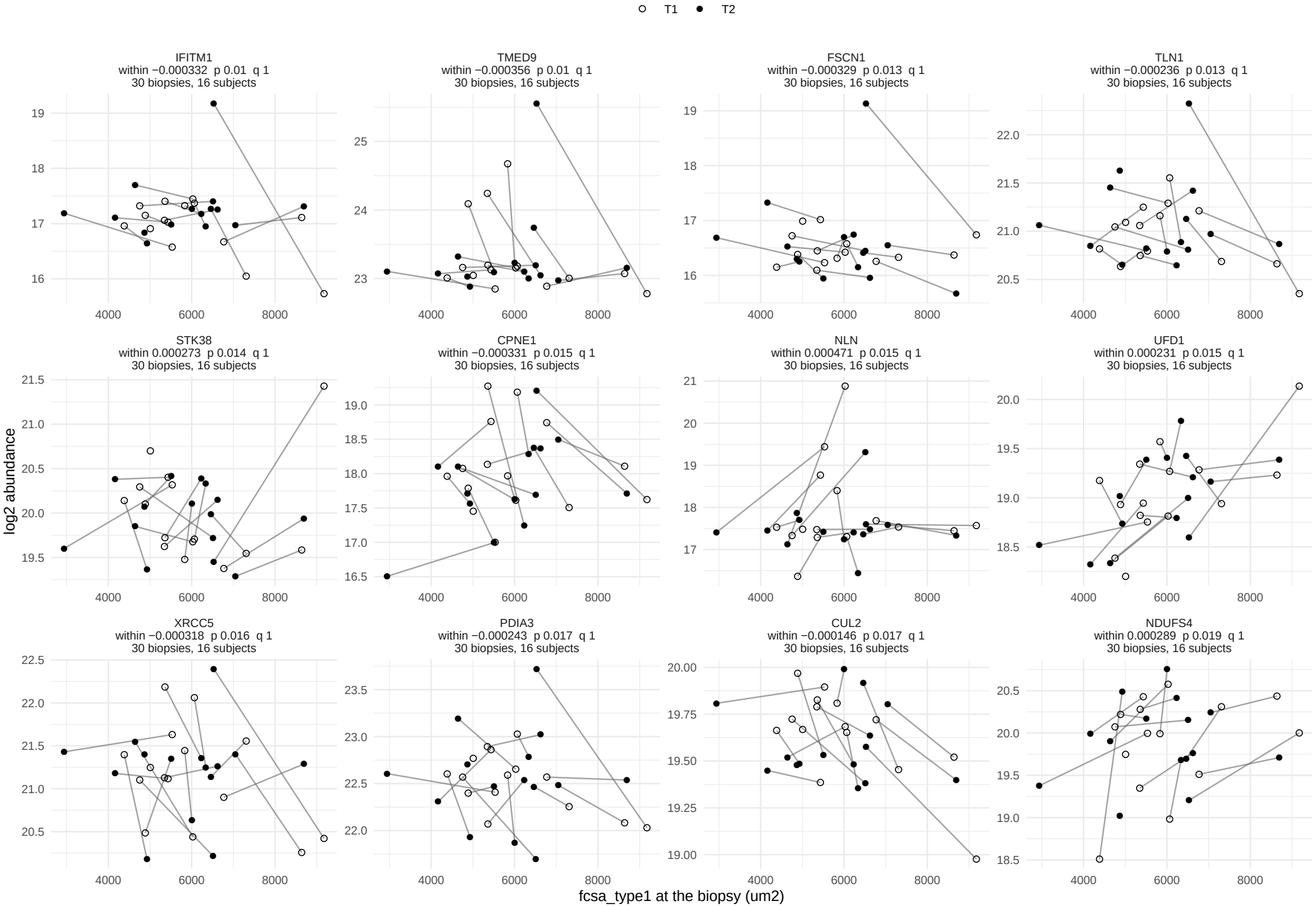
68 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type1, within−person term

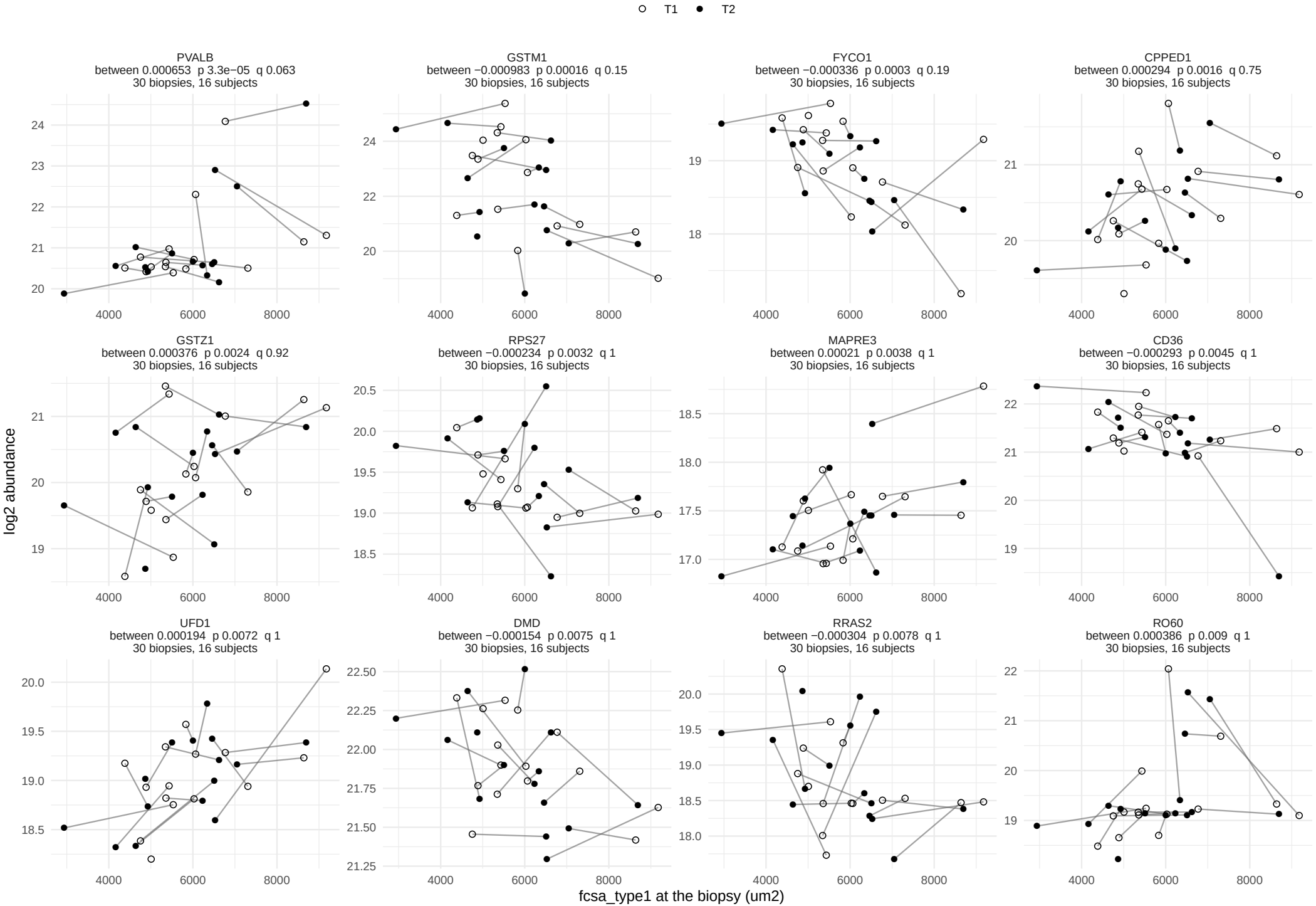
68 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type1, between−person term

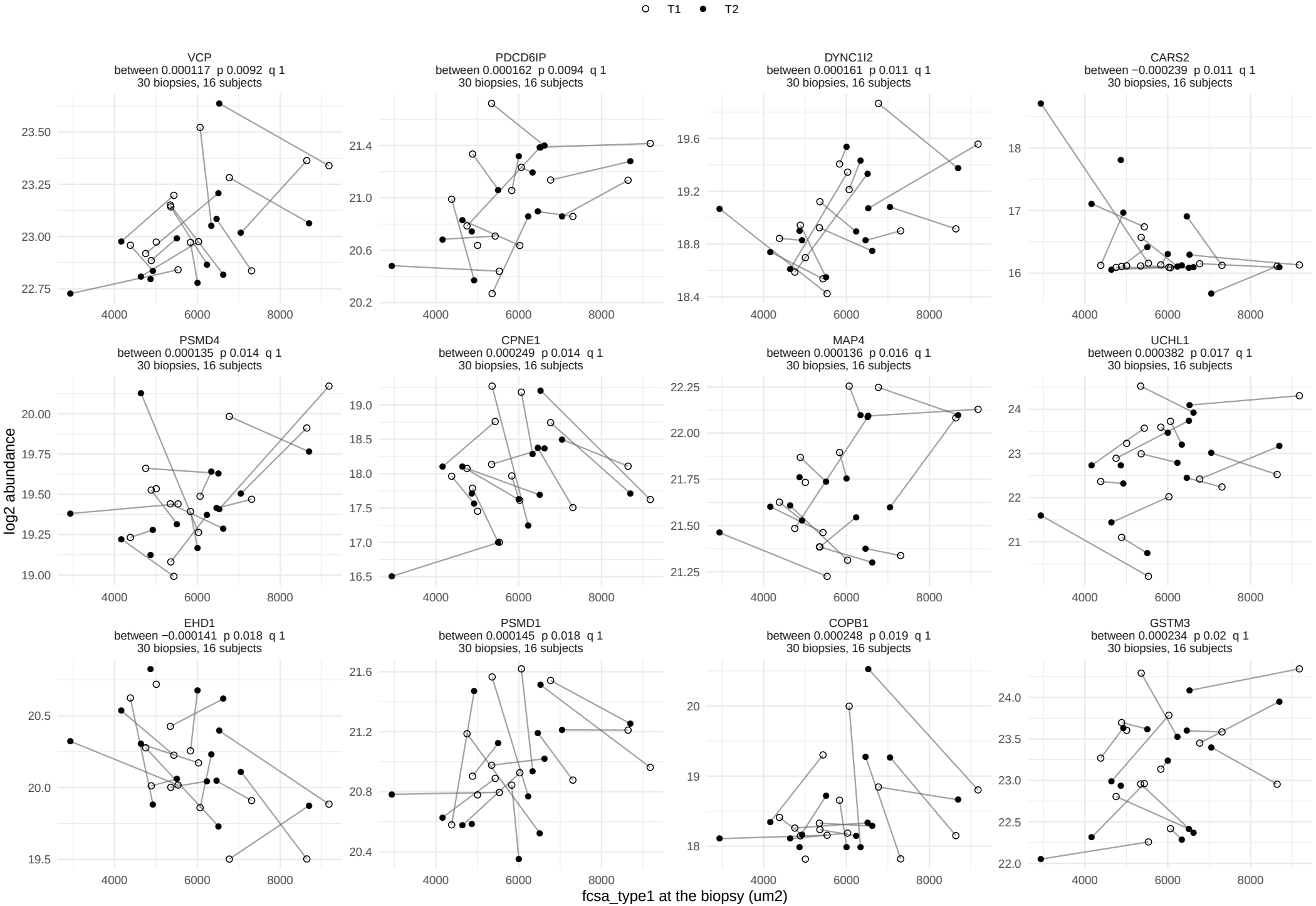
69 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type1, between−person term

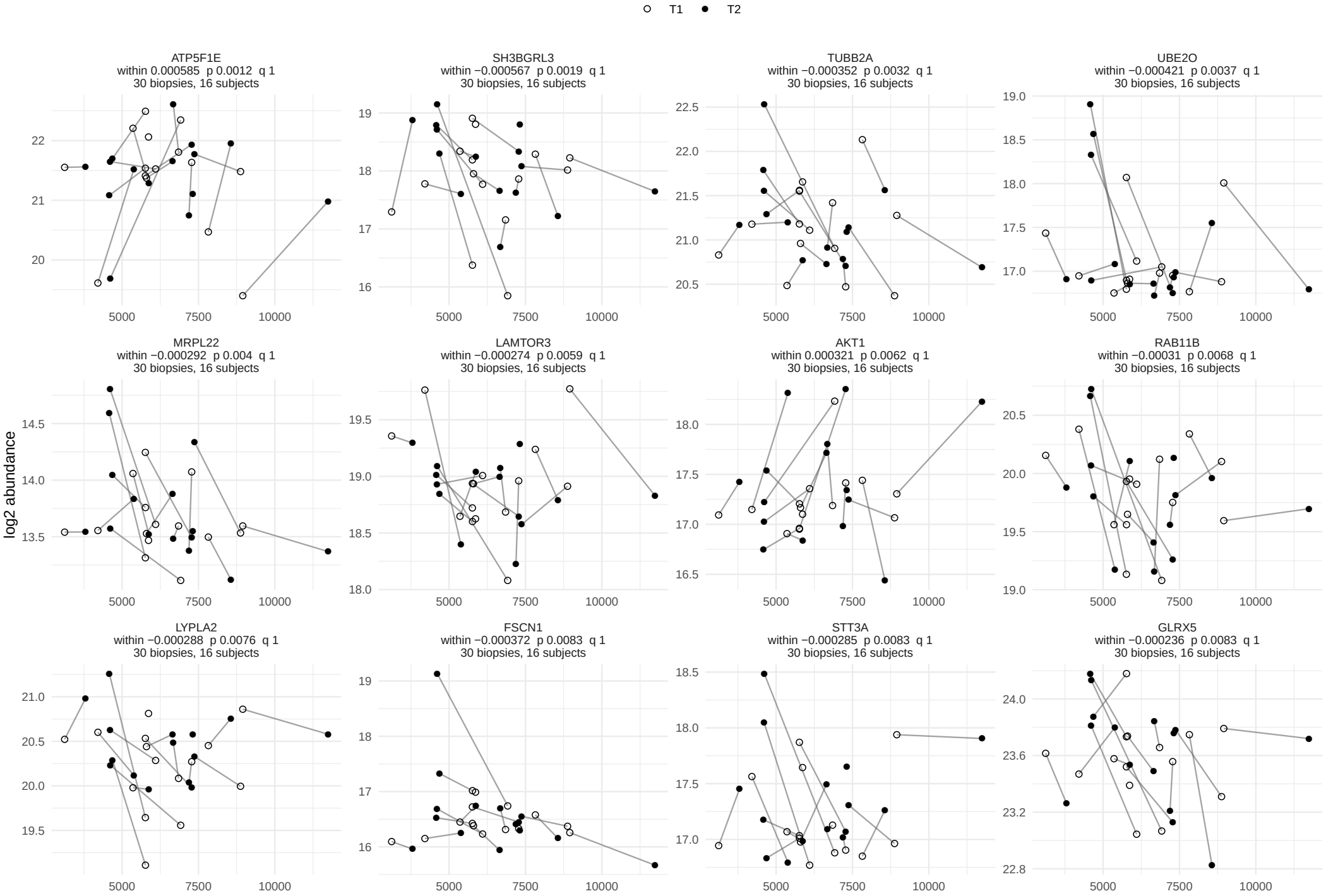
69 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type2, within−person term

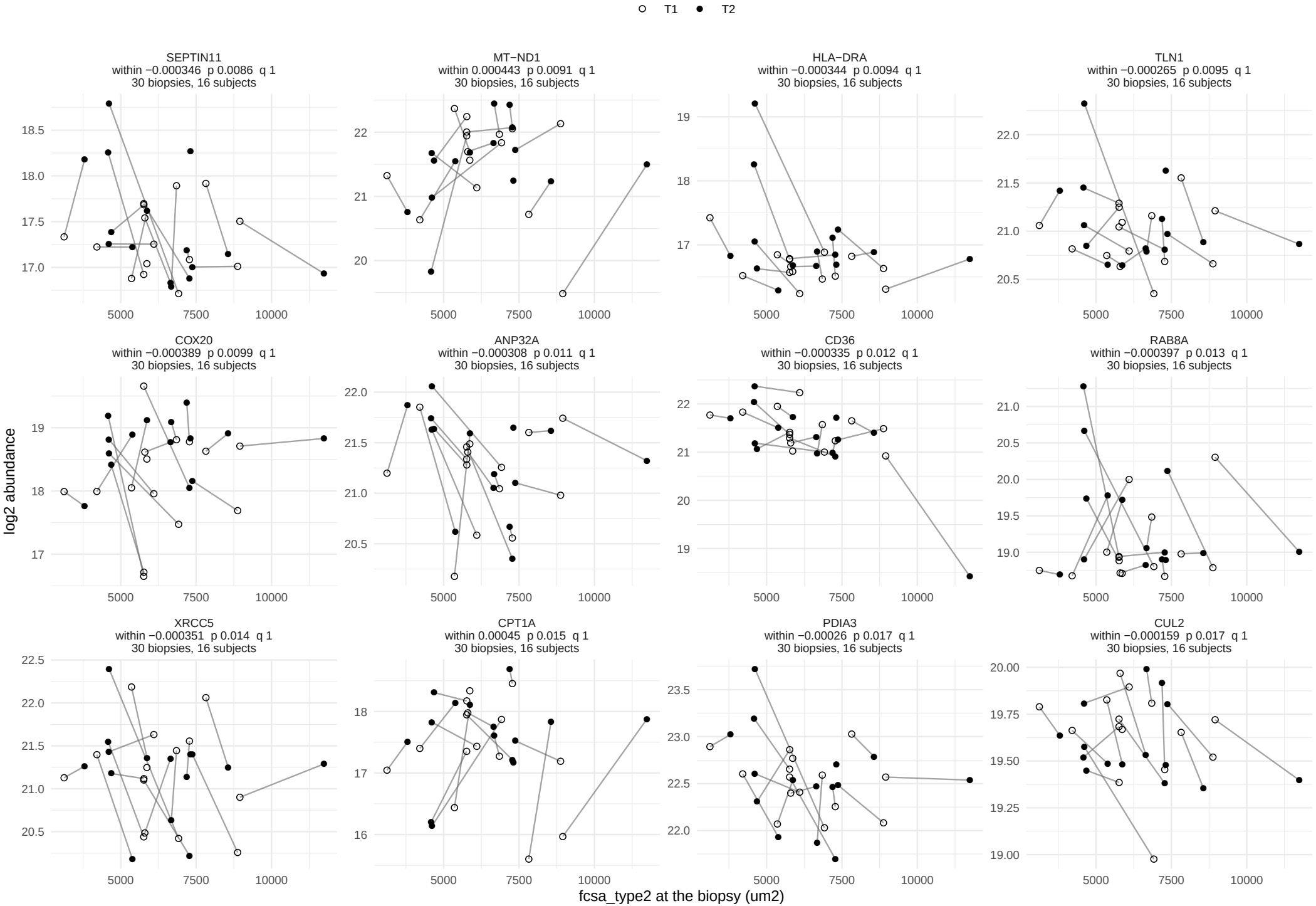
76 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type2, within−person term

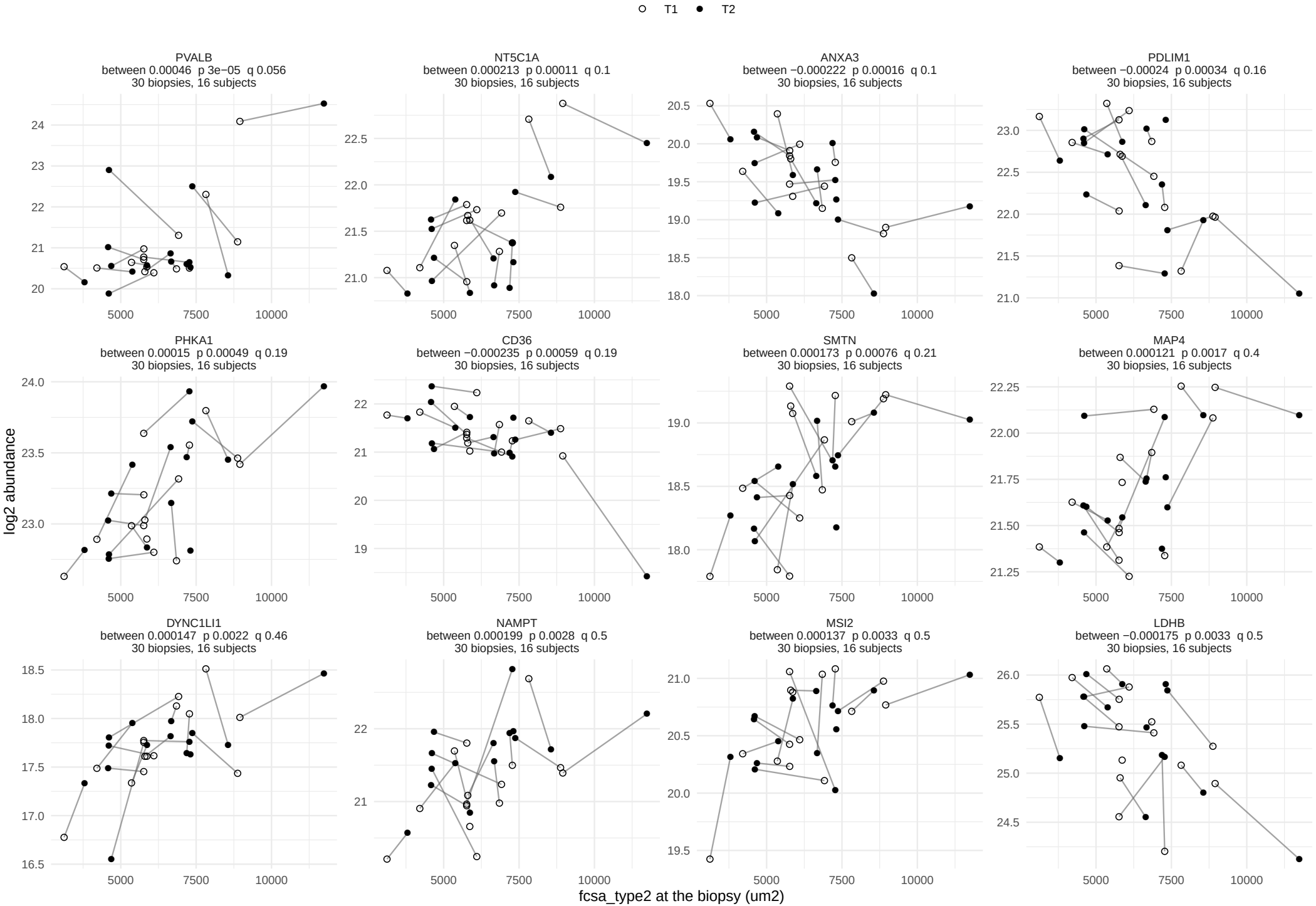
76 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type2, between−person term

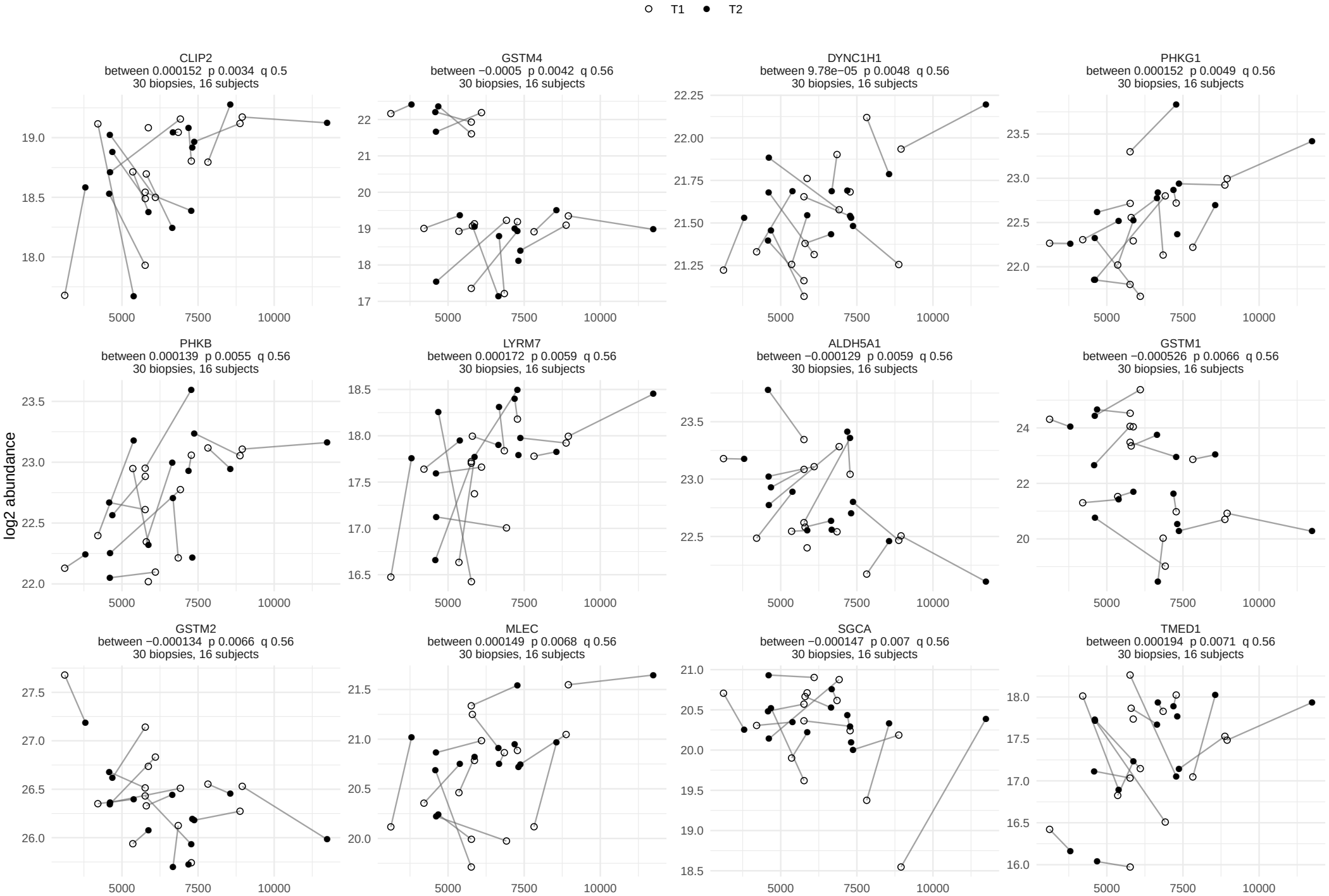
86 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

fcsa_type2, between−person term

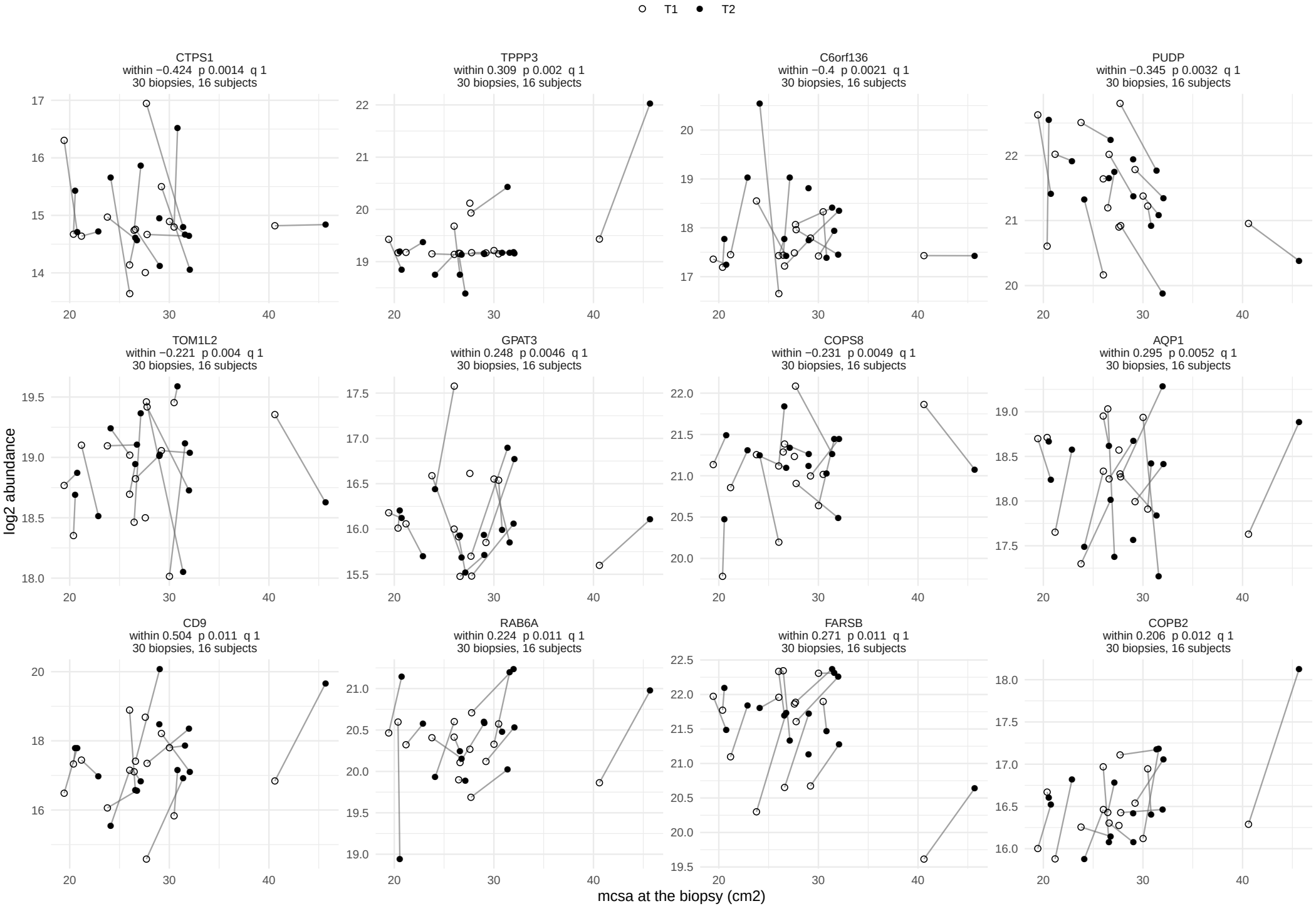
86 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

mcsa, within−person term

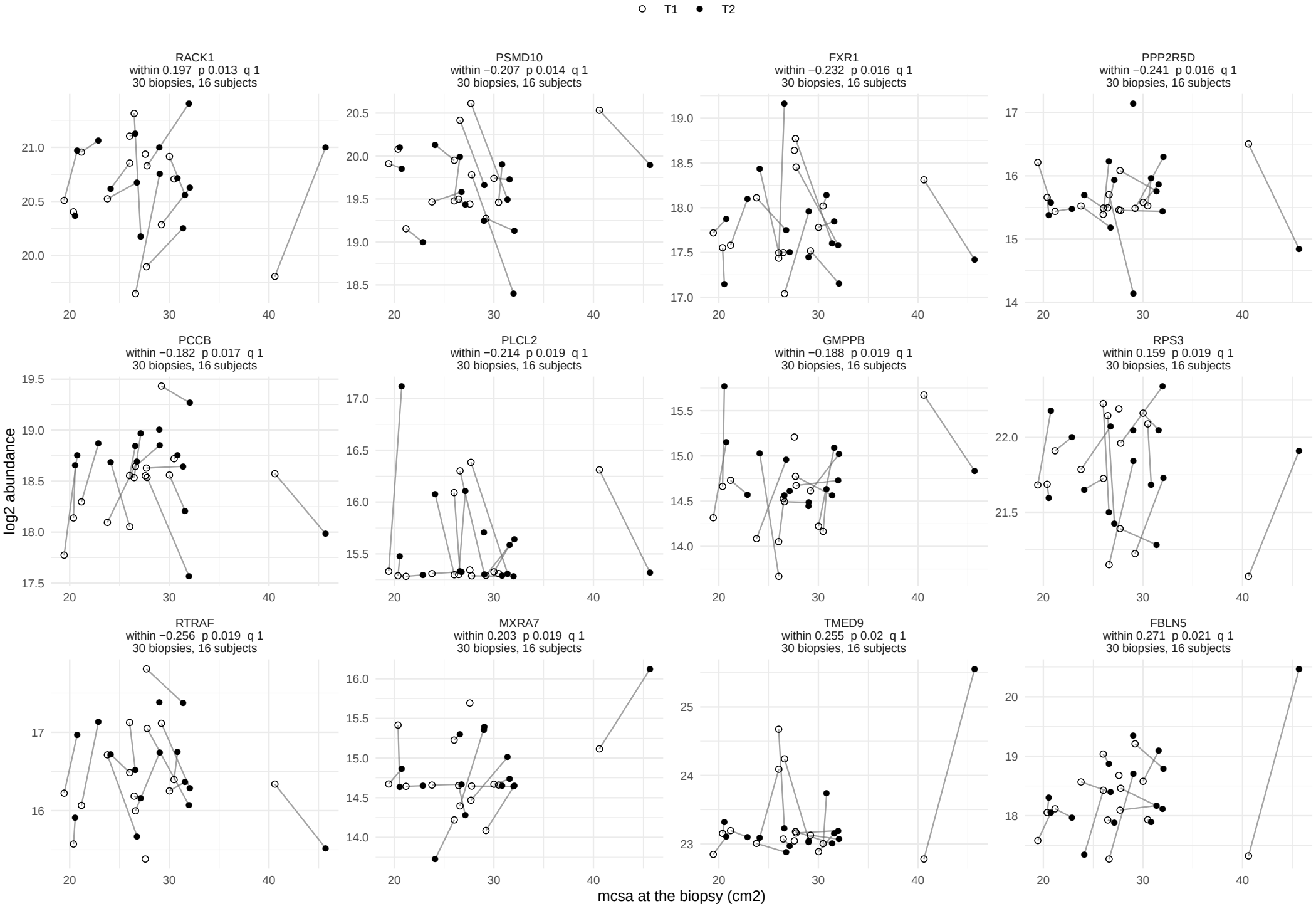
62 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

mcsa, within−person term

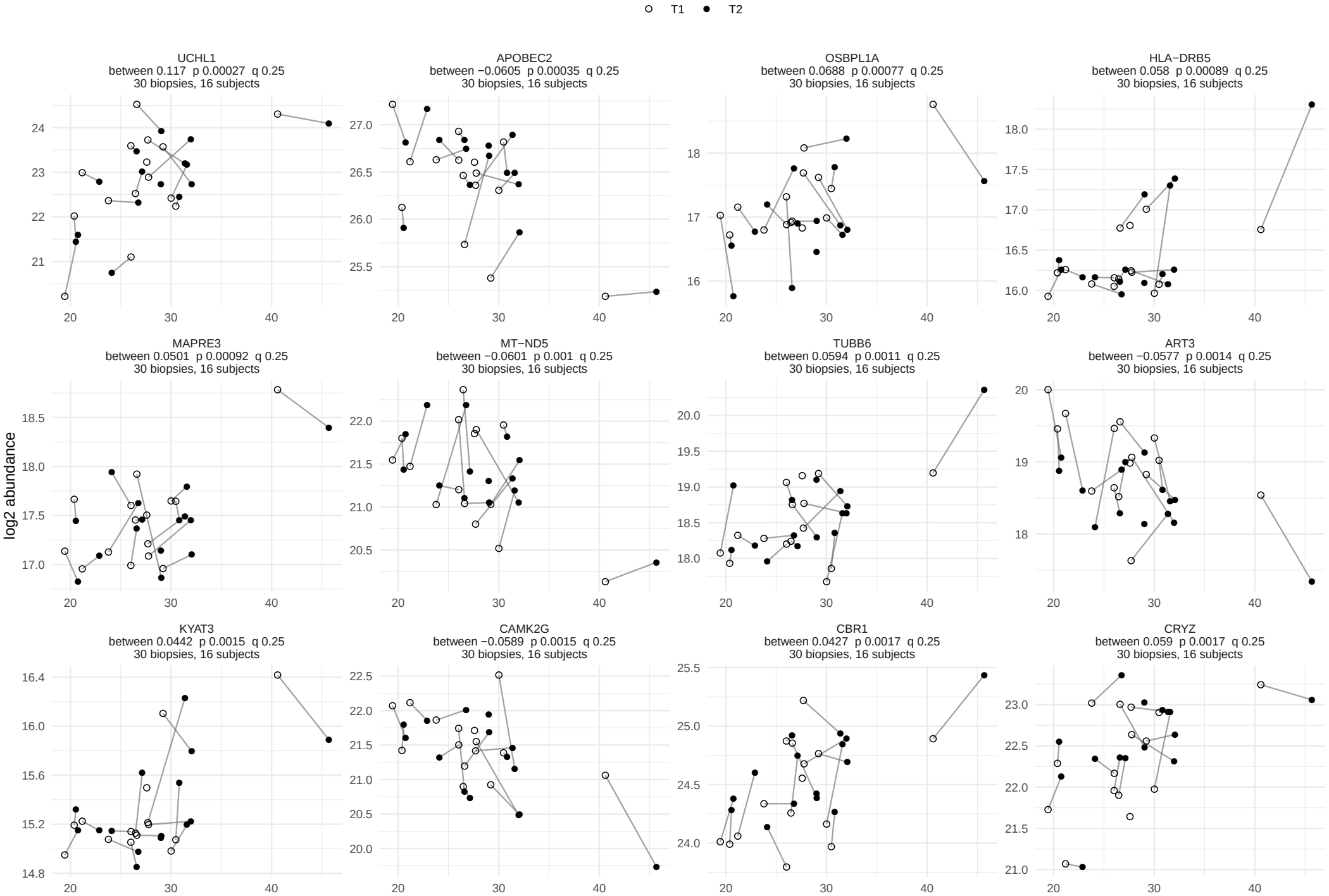
62 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

mcsa, between−person term

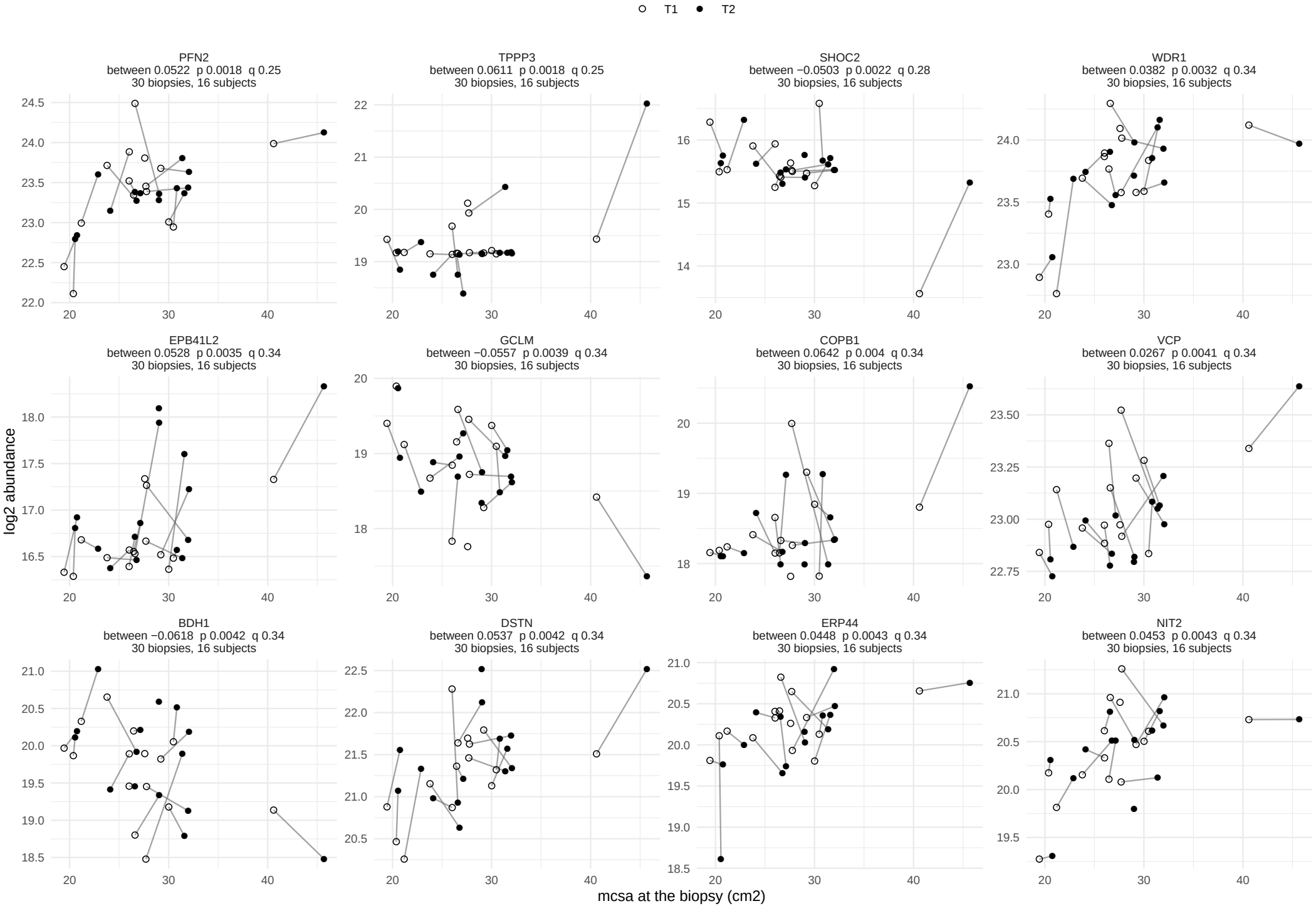
179 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

mcsa, between−person term

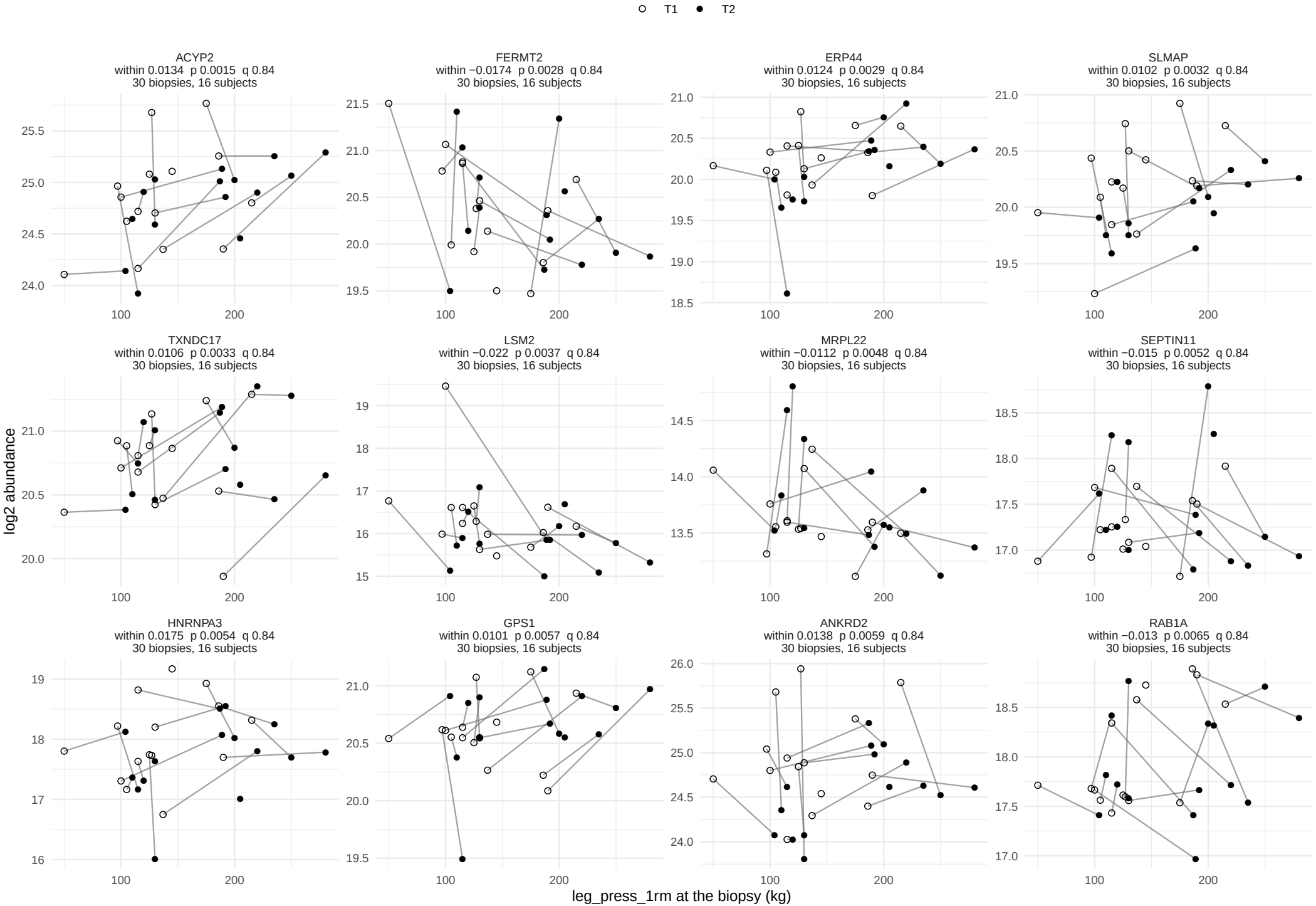
179 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_press_1rm, within−person term

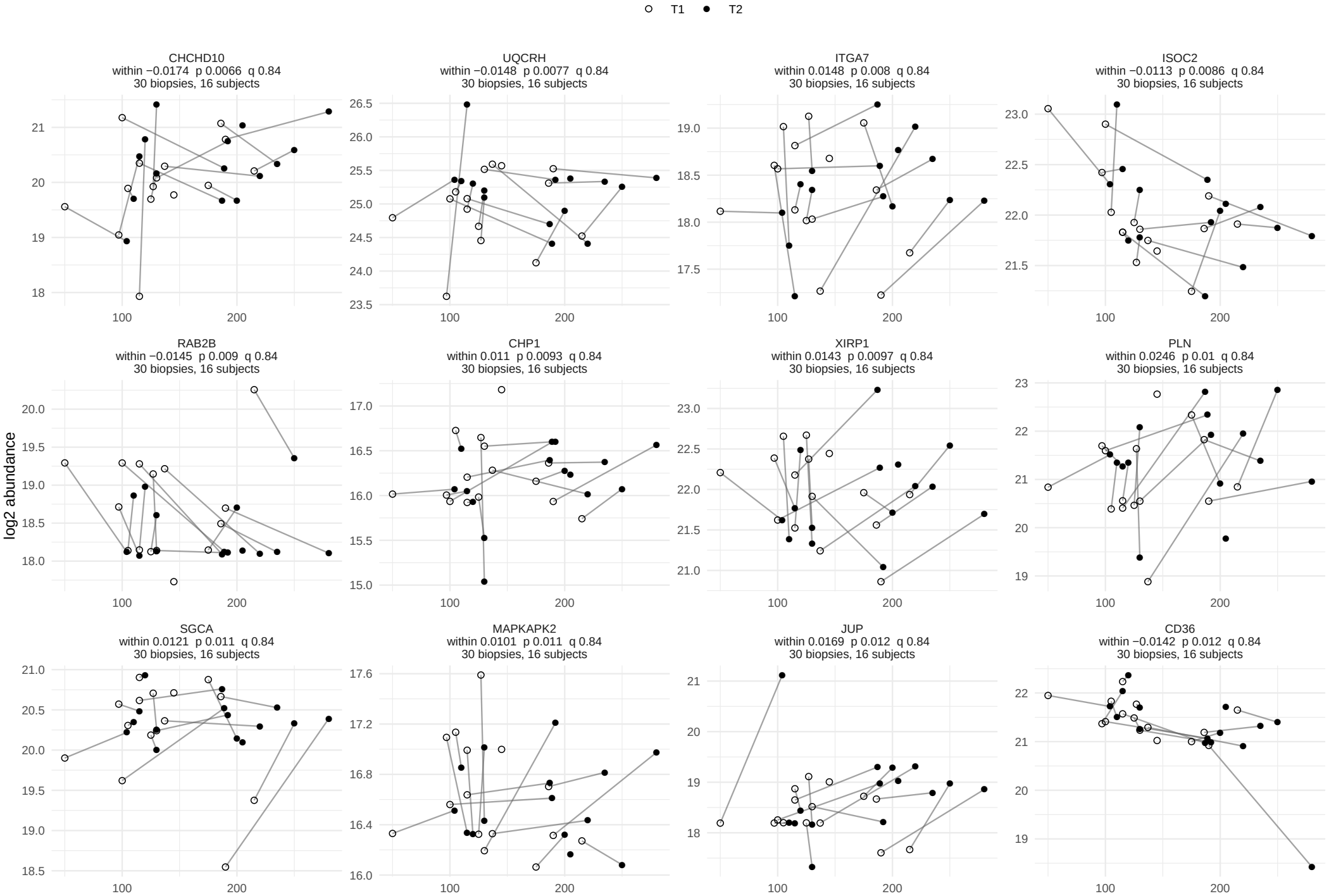
103 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_press_1rm, within−person term

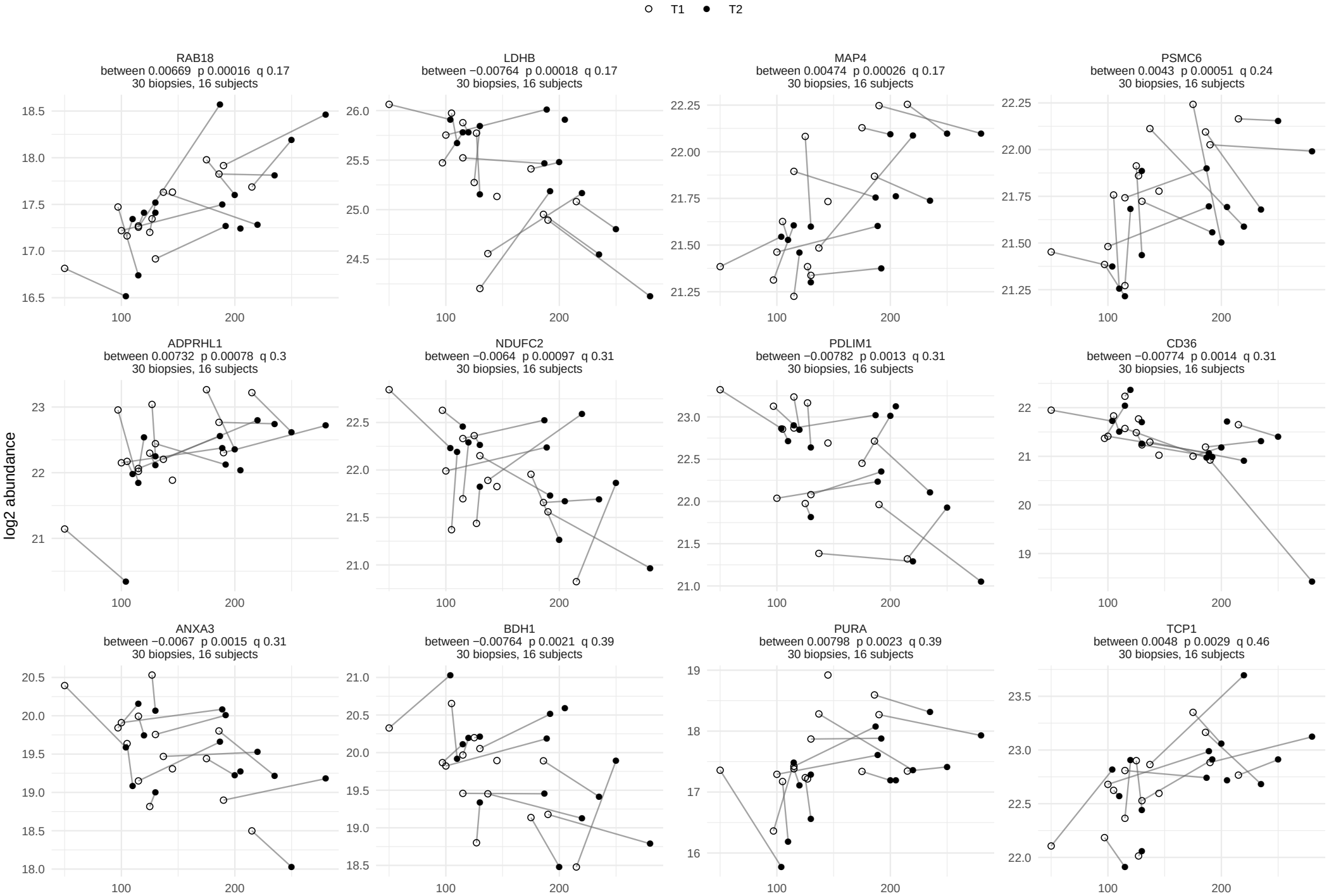
103 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_press_1rm, between−person term

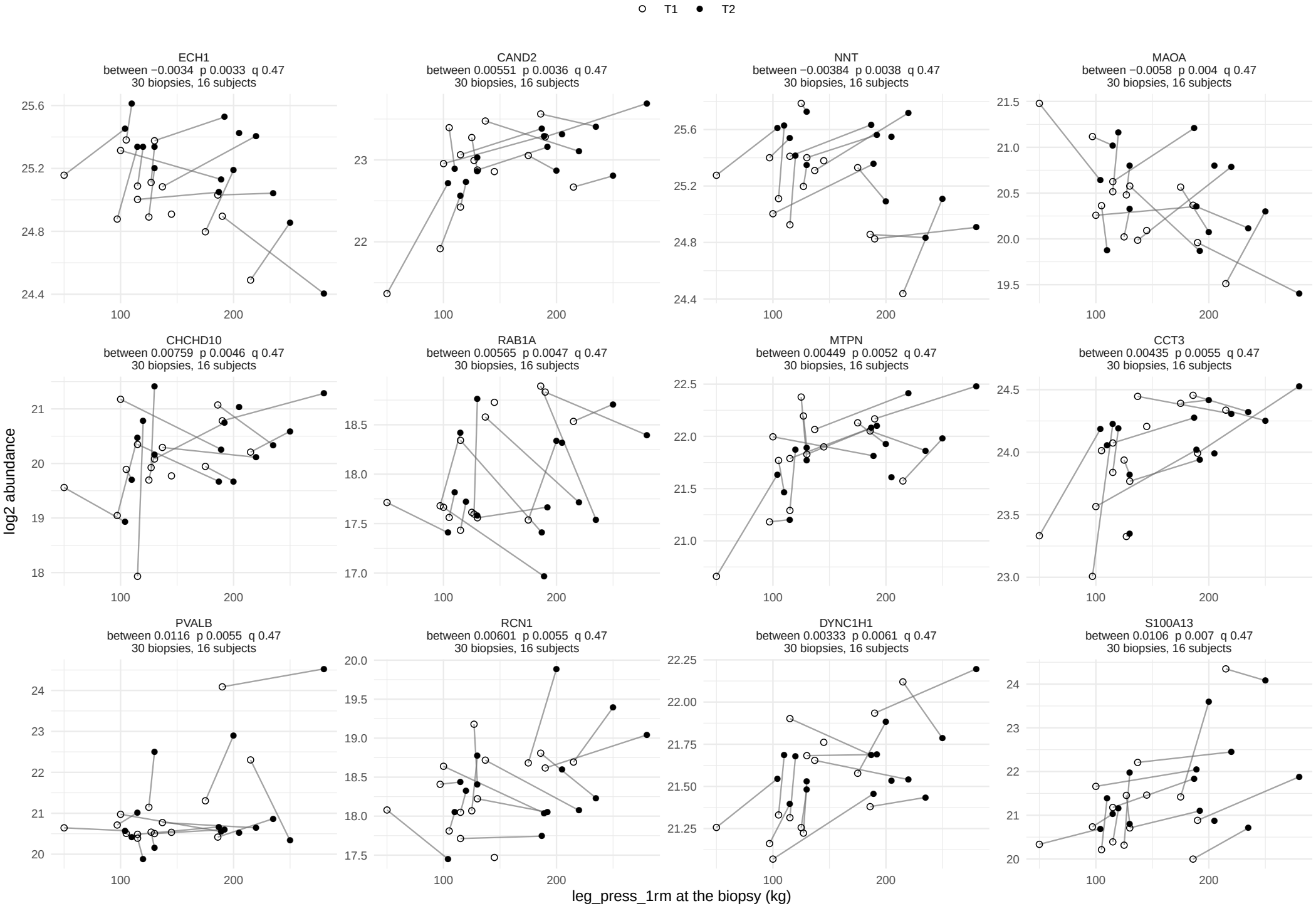
132 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_press_1rm, between−person term

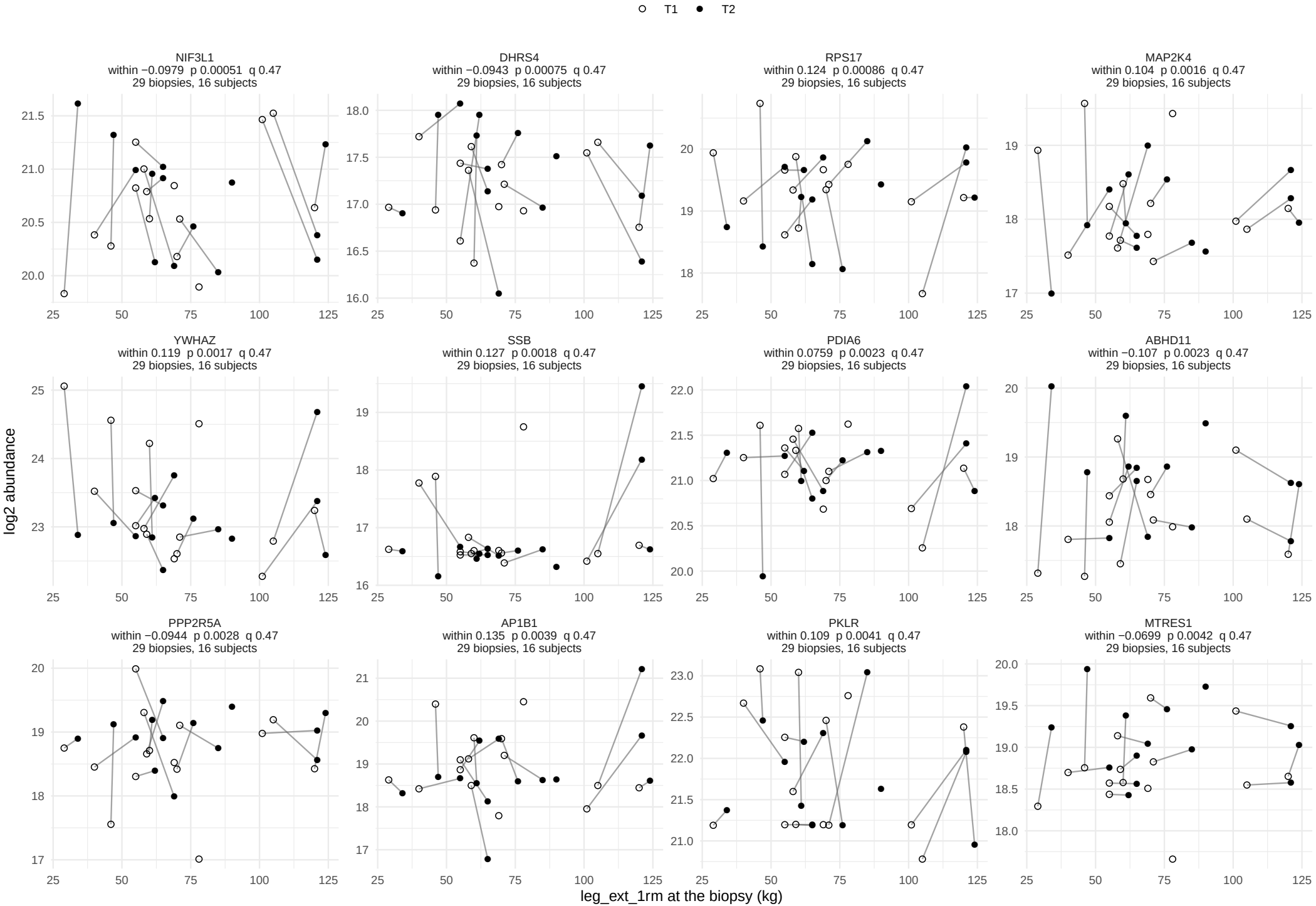
132 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_ext_1rm, within−person term

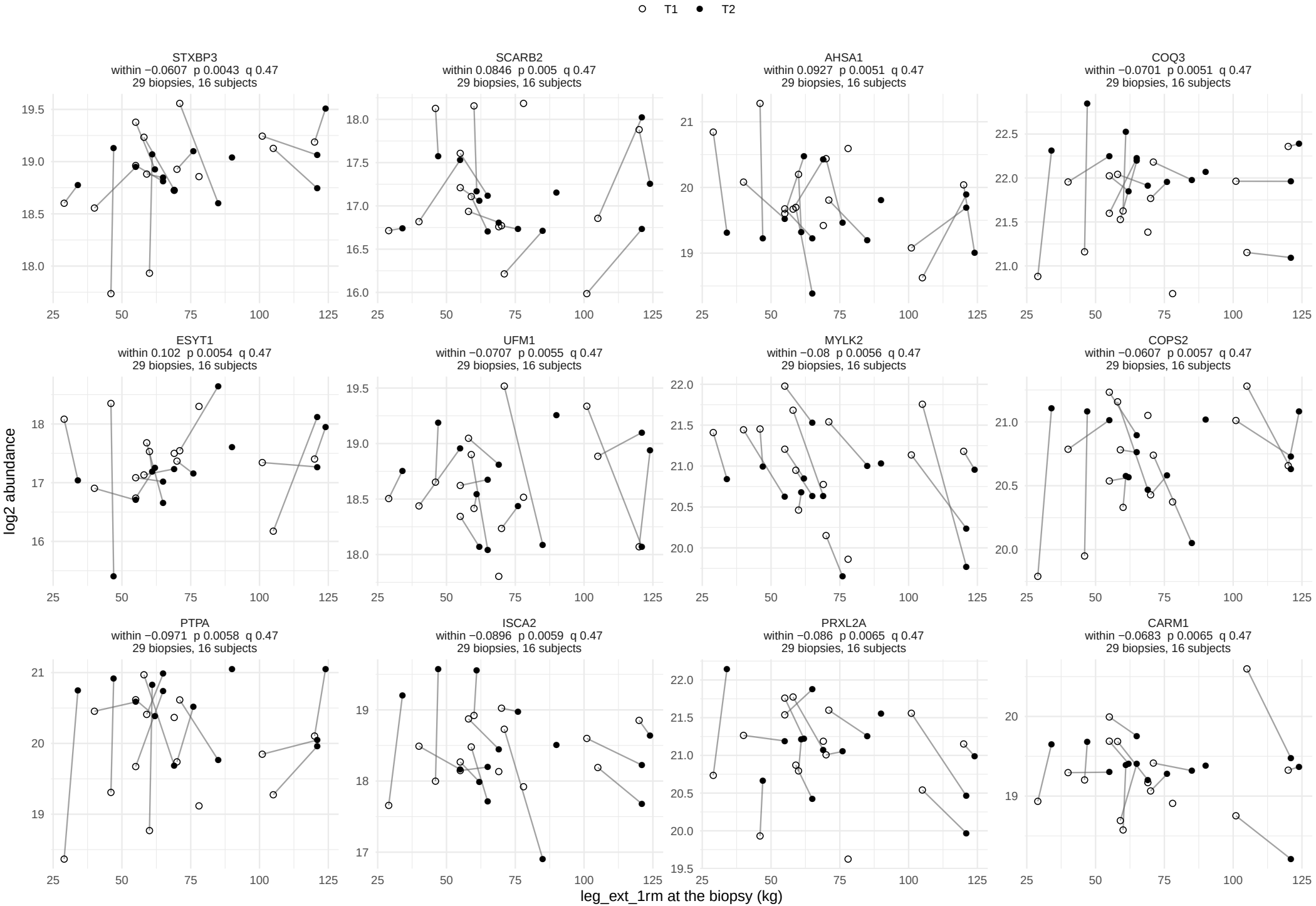
134 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_ext_1rm, within−person term

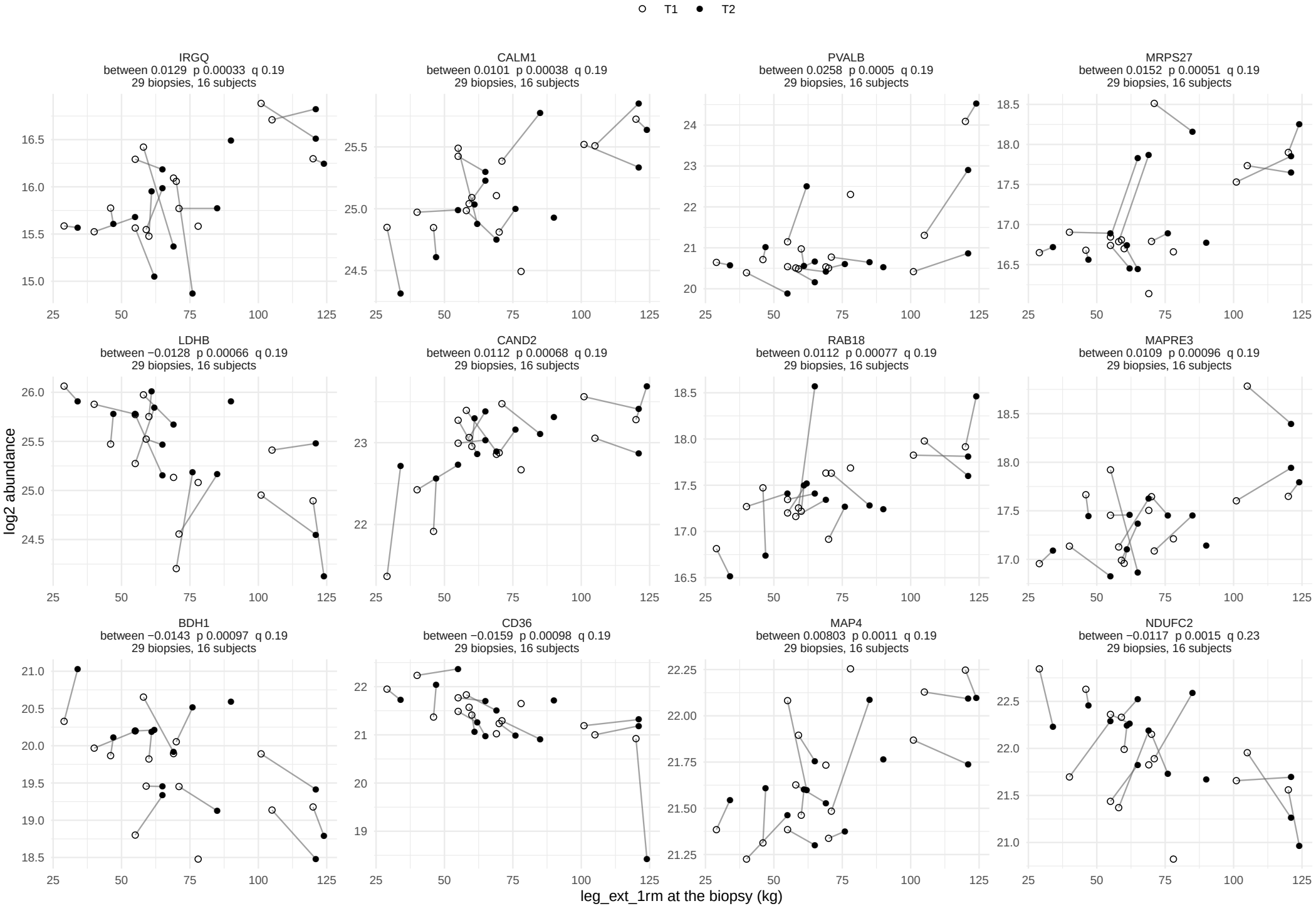
134 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_ext_1rm, between−person term

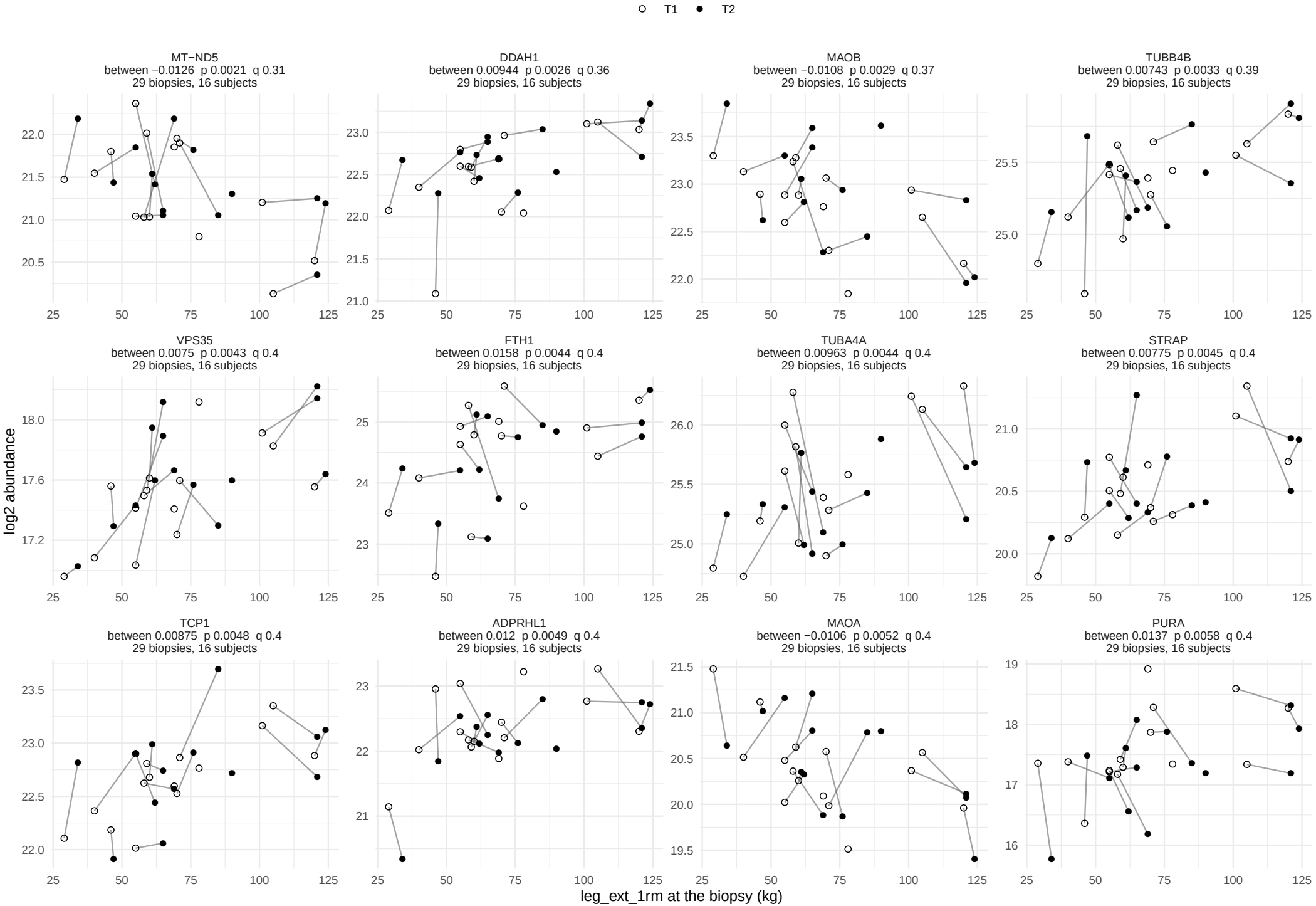
130 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

leg_ext_1rm, between−person term

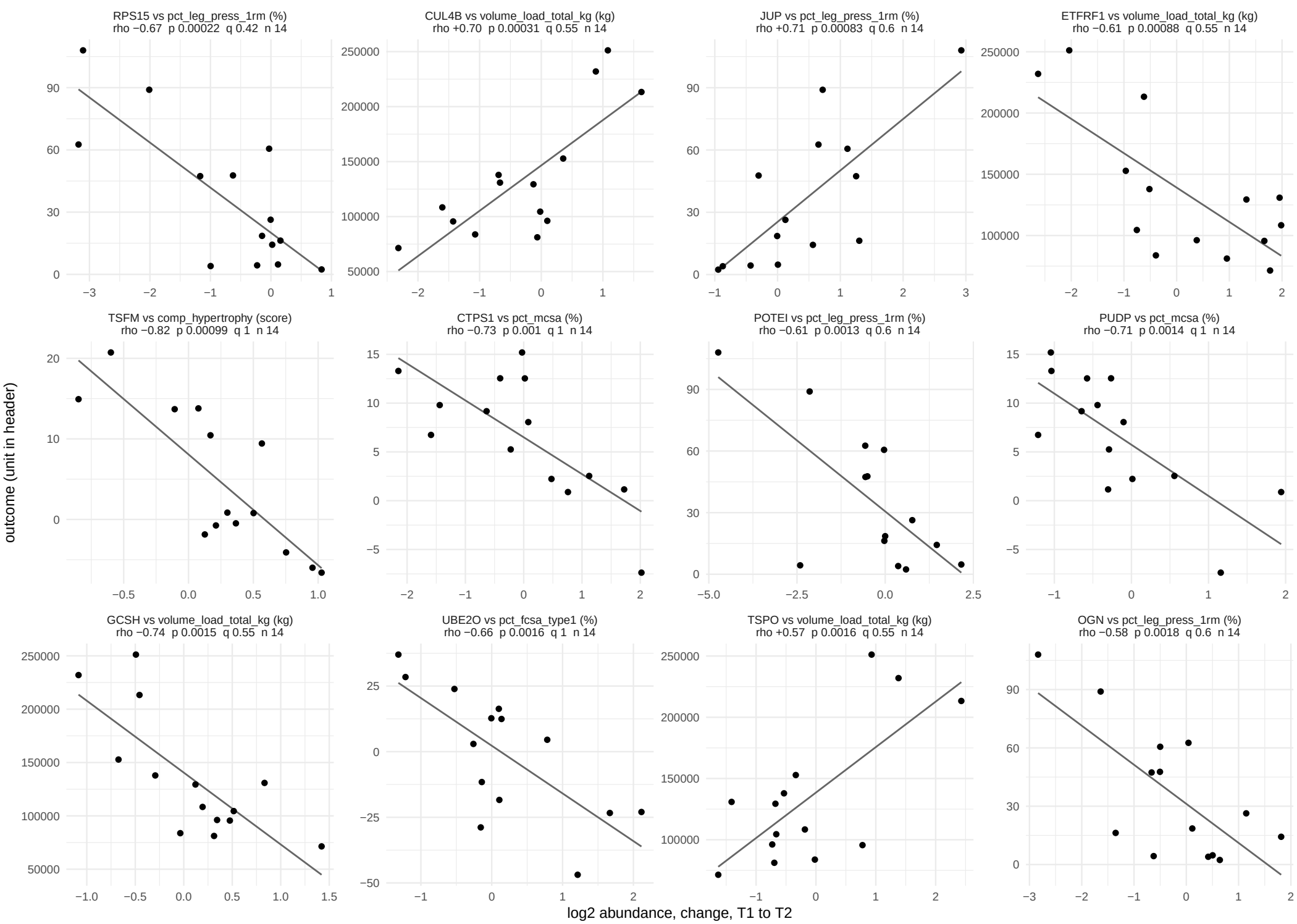
130 of 1900 proteins at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



limma, abundance ~ timepoint + between + within, subject blocked, on the T1 and T2 biopsies with the trait measured. Lines join each subject's two biopsies: the within term reads their slopes, the between term the spread of subjects. Header: effect per trait unit, p, BH q within trait and term, and the biopsies and subjects where the feature was observed. Table: c_data/01_proteins.xlsx, model.

Change score, training window

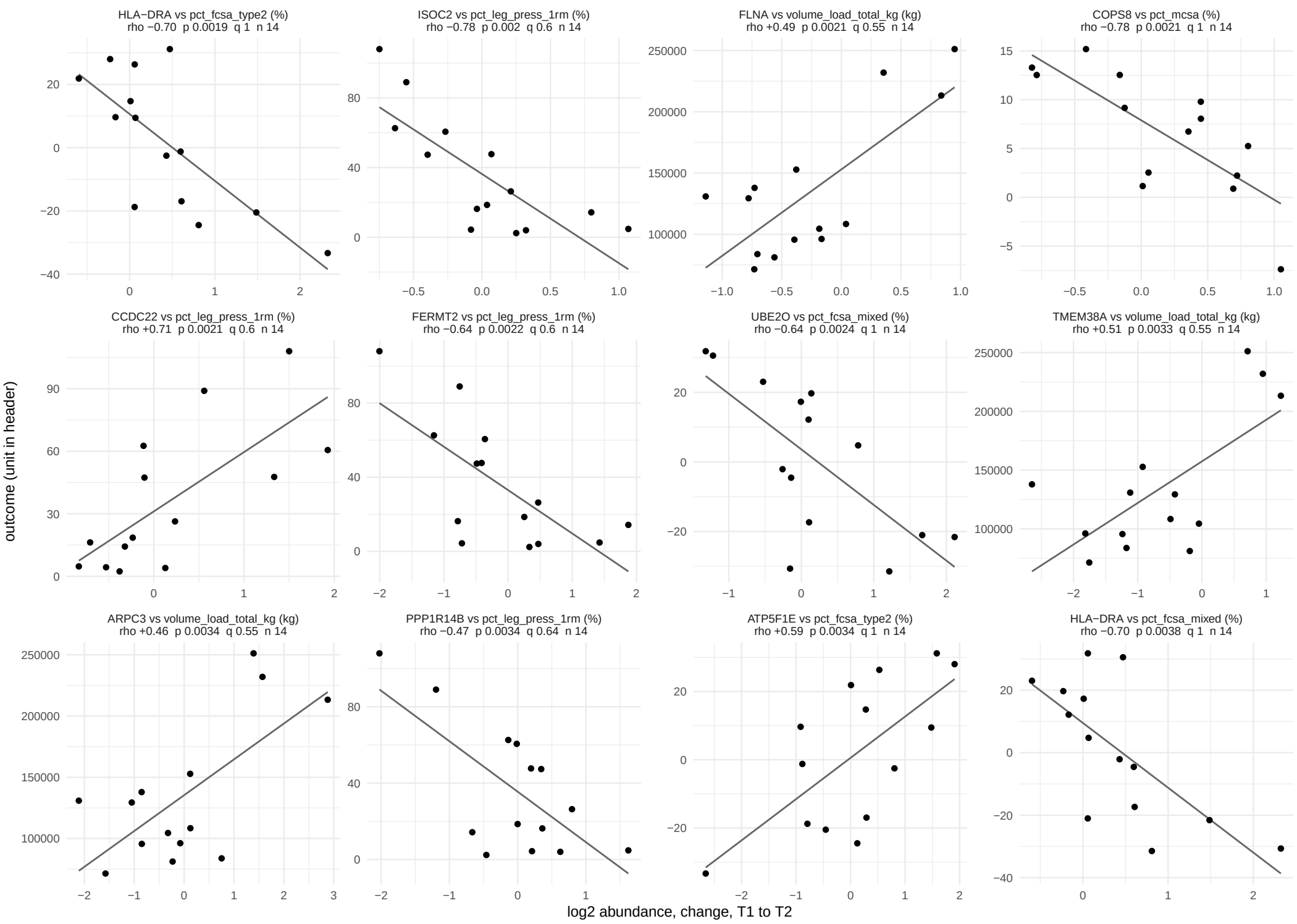
605 of 15200 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, training window

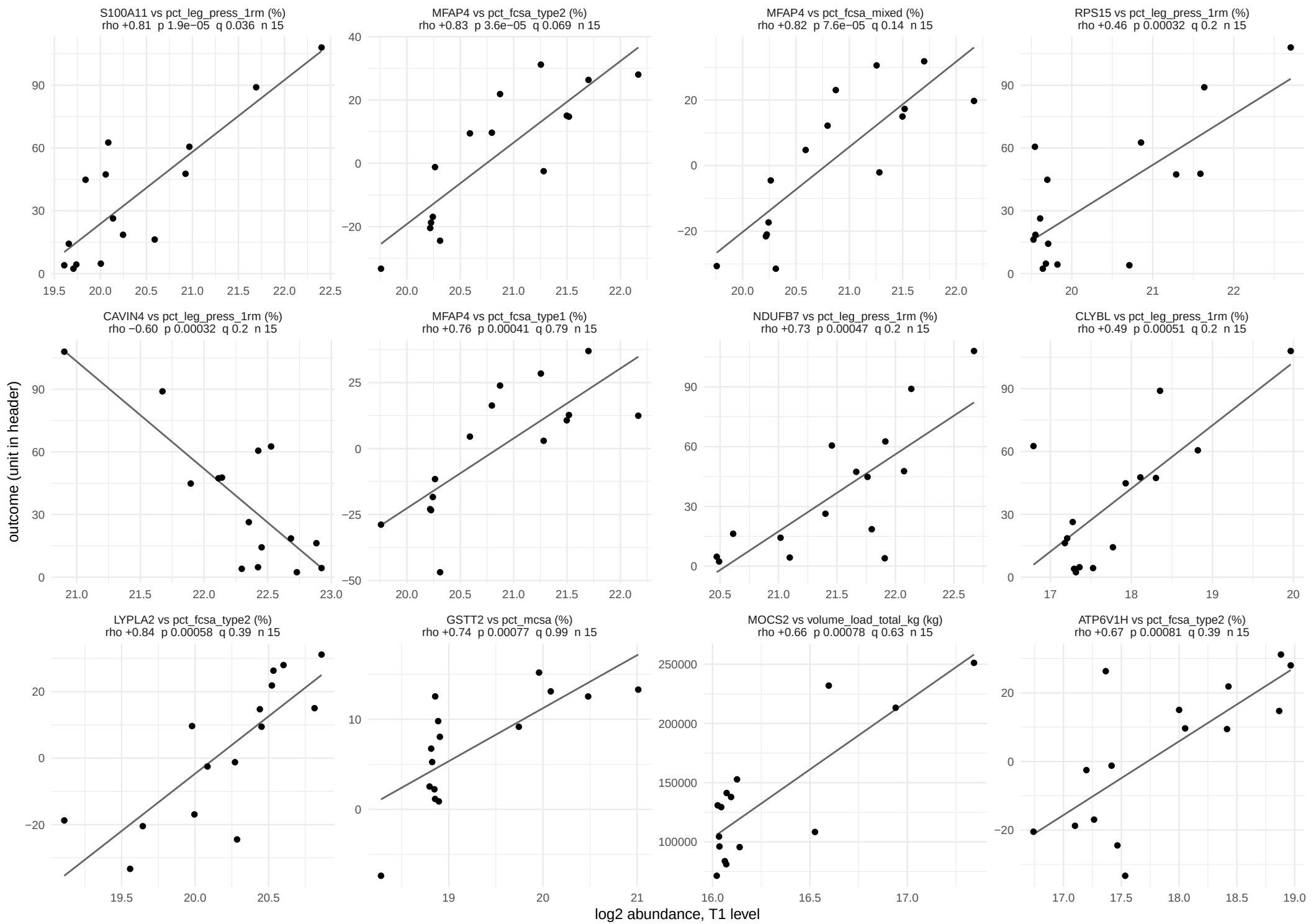
605 of 15200 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, baseline window

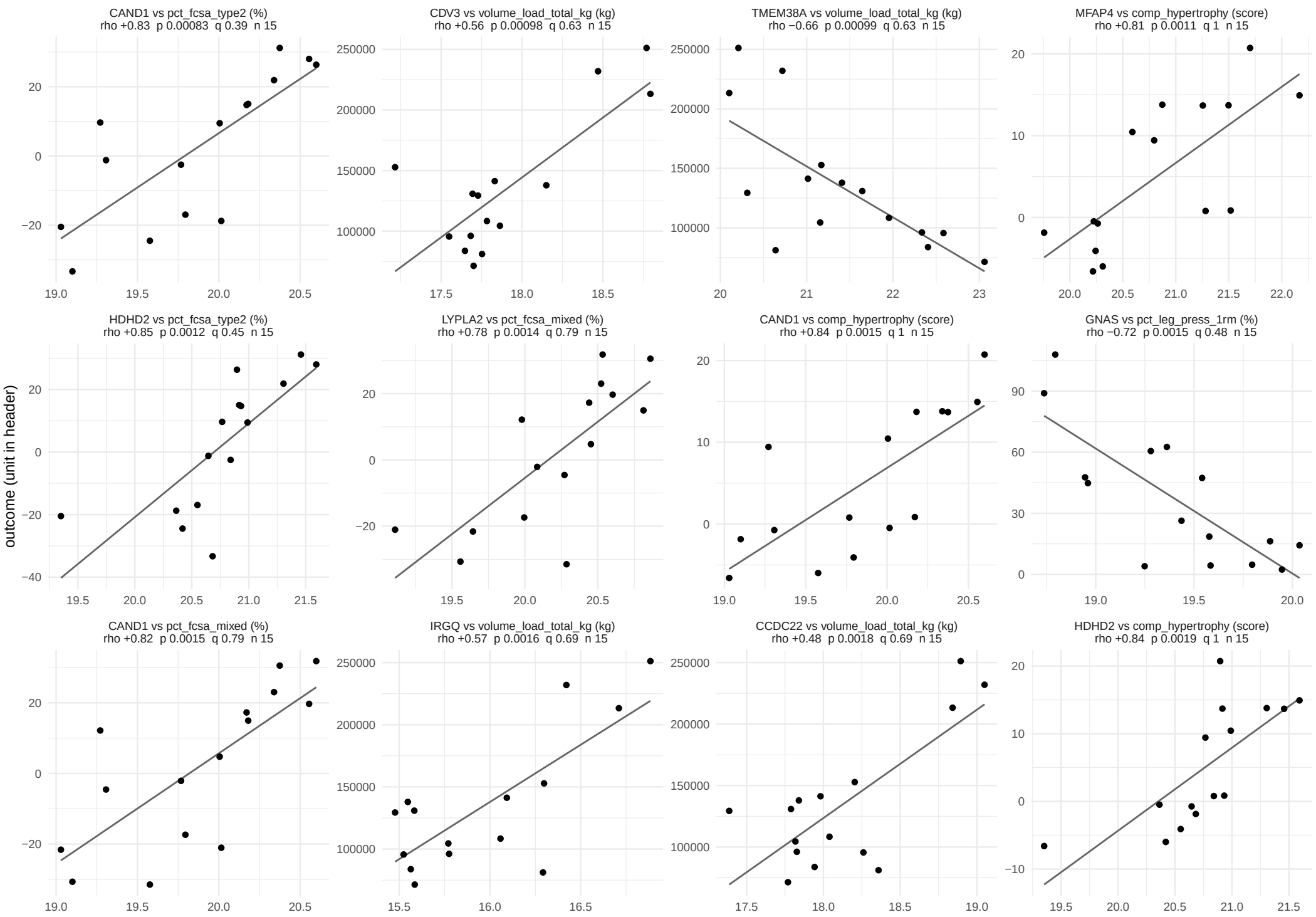
686 of 15200 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, baseline window

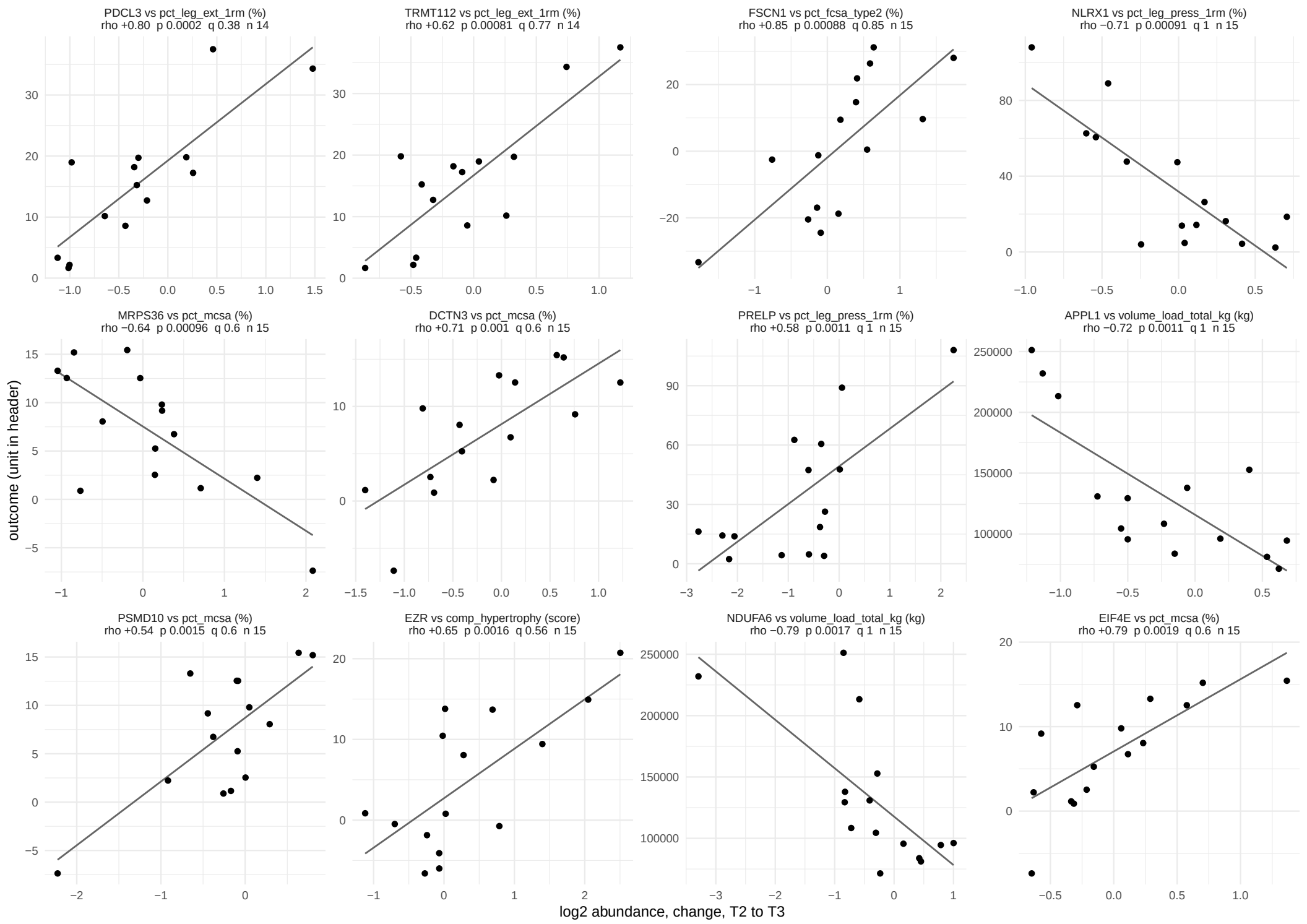
686 of 15200 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, acute window

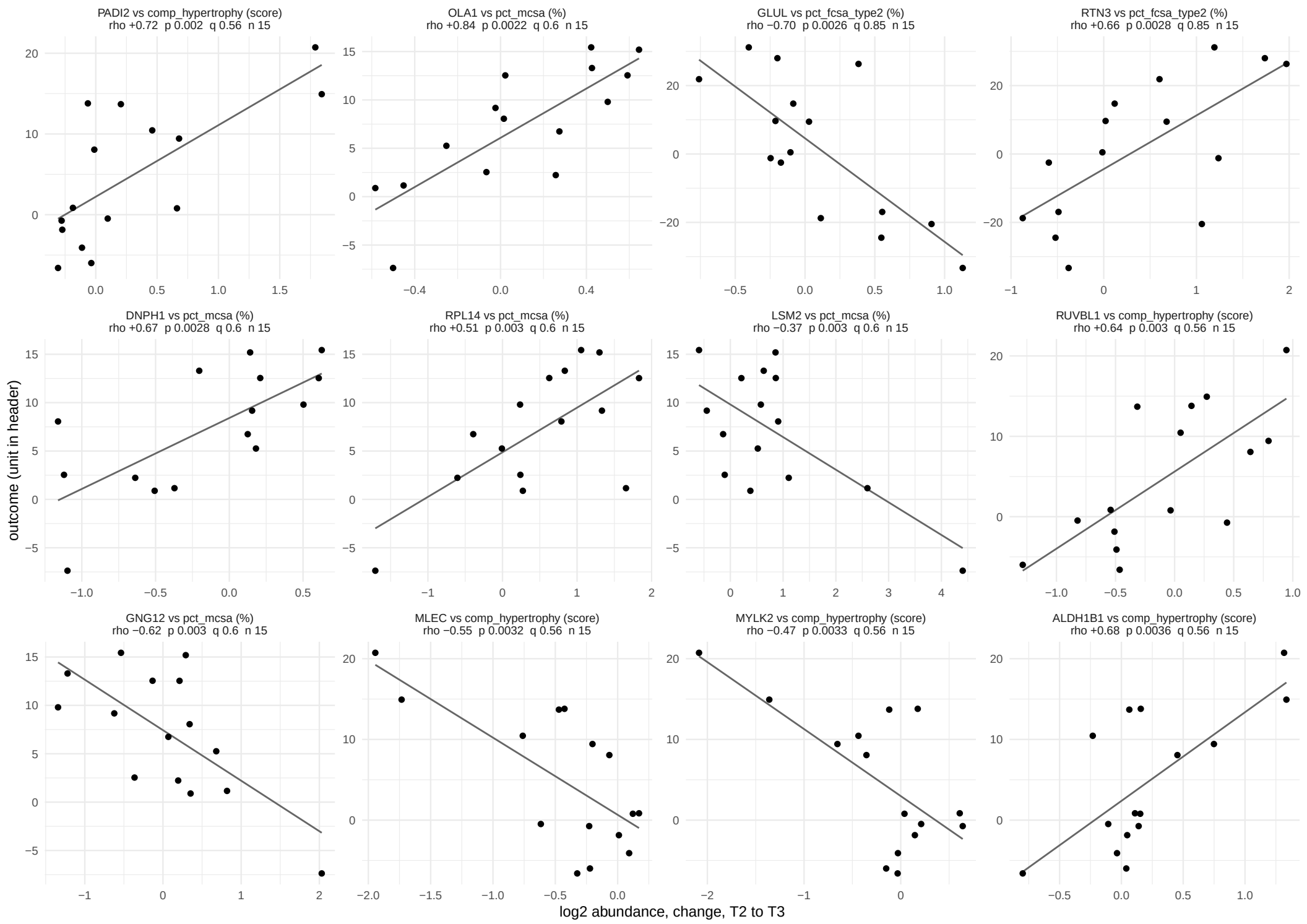
629 of 15200 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, acute window

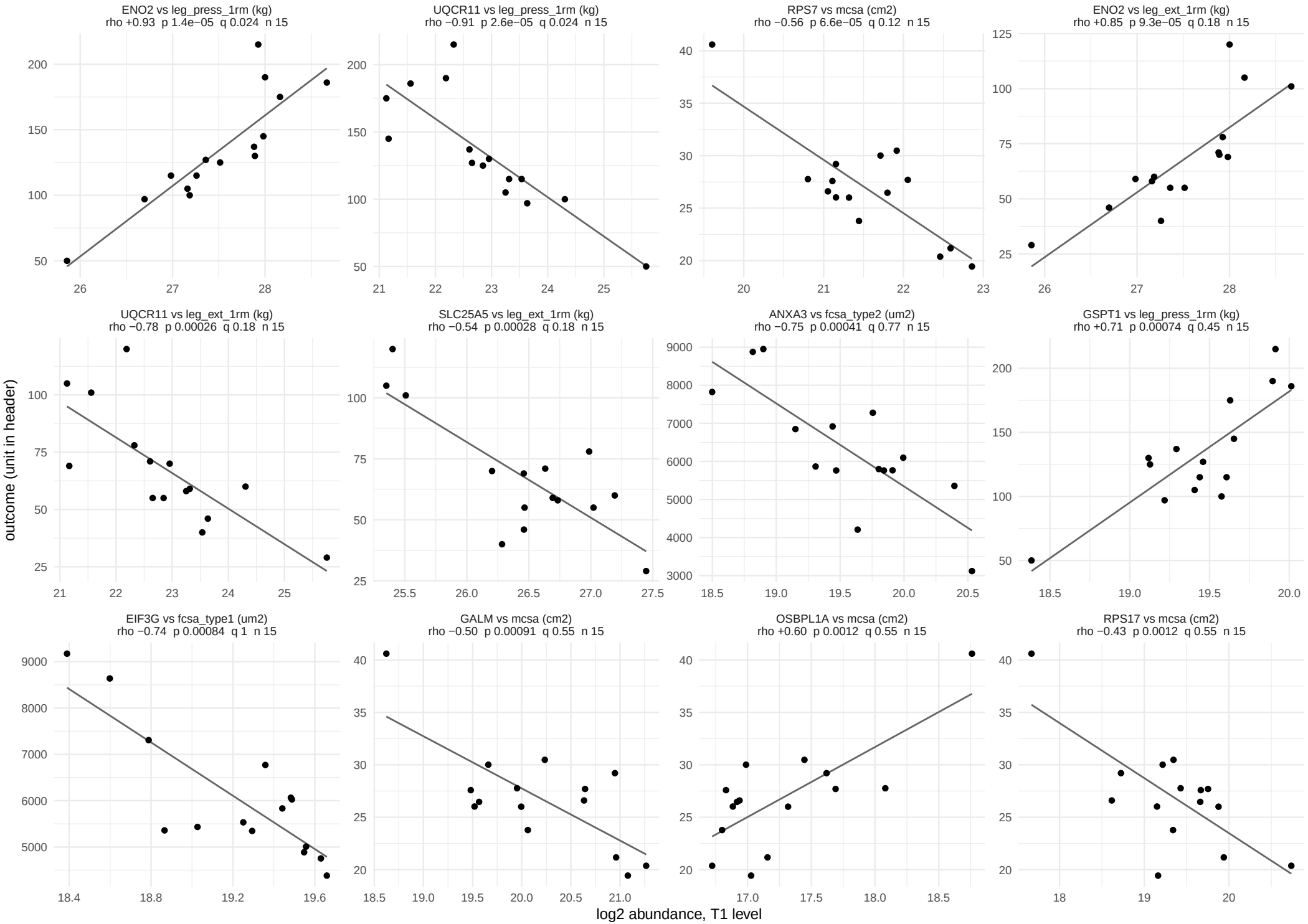
629 of 15200 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, baseline_level window

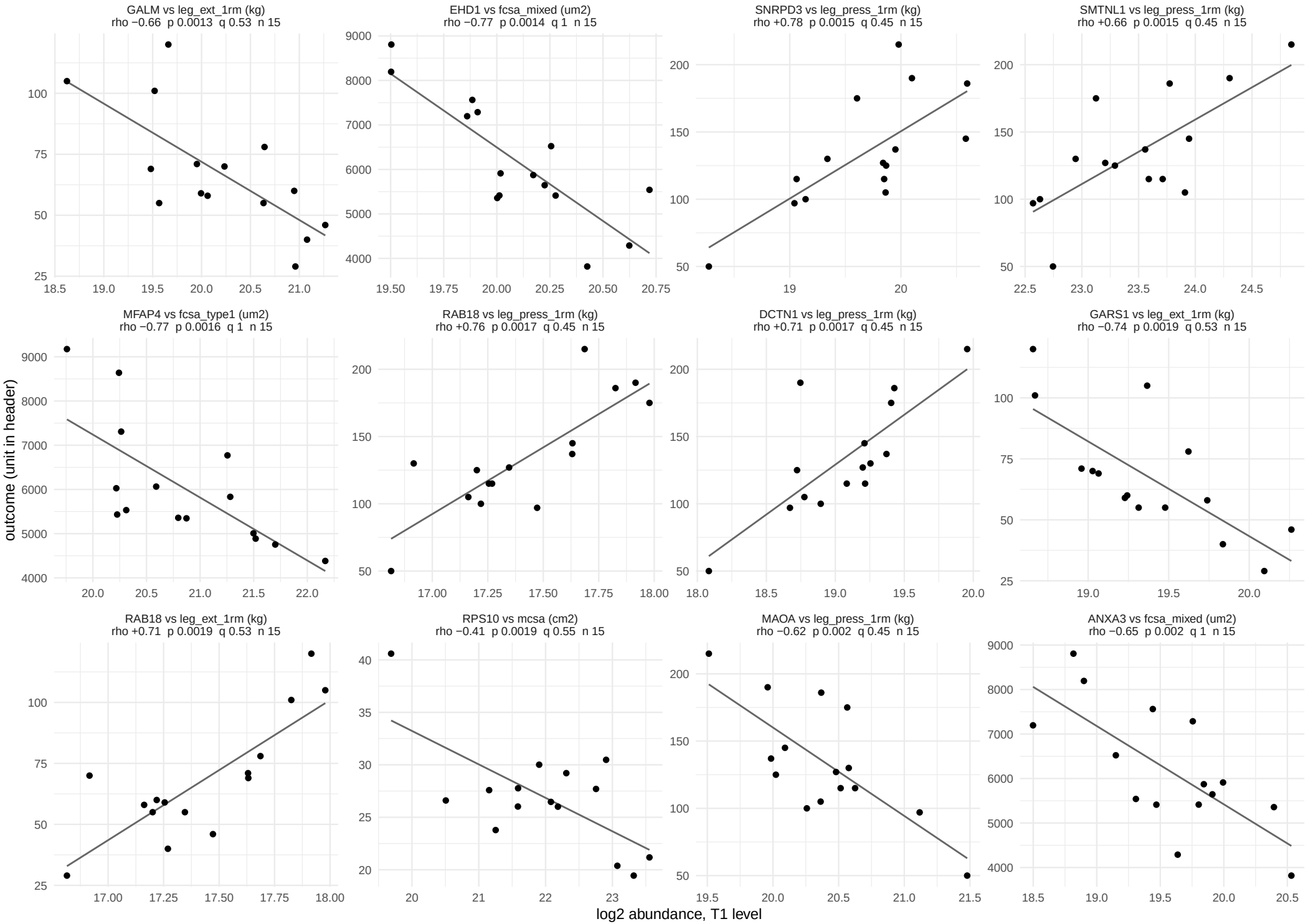
704 of 11400 feature−outcome pairs at p < 0.05, 2 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, baseline_level window

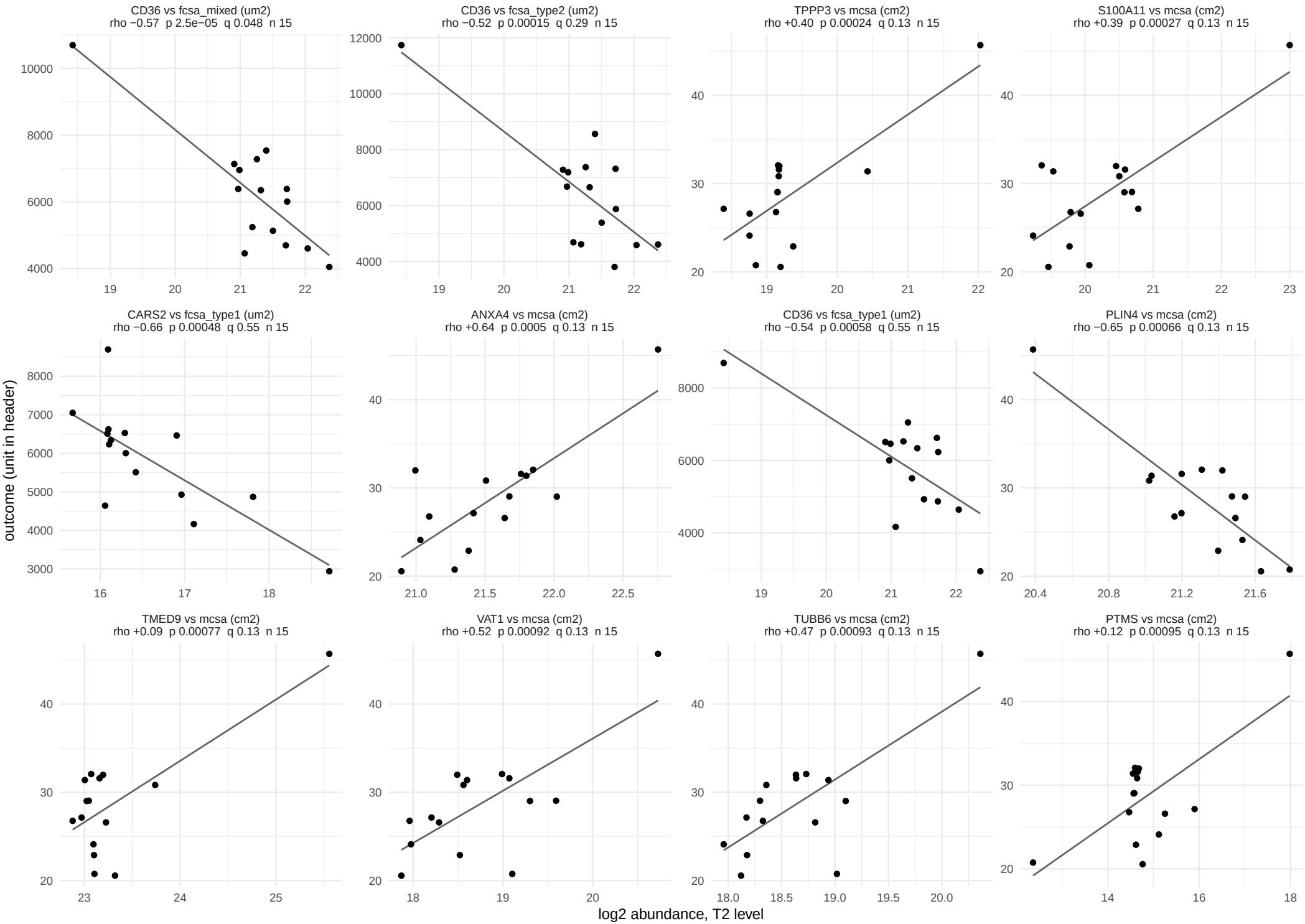
704 of 11400 feature−outcome pairs at p < 0.05, 2 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, post_level window

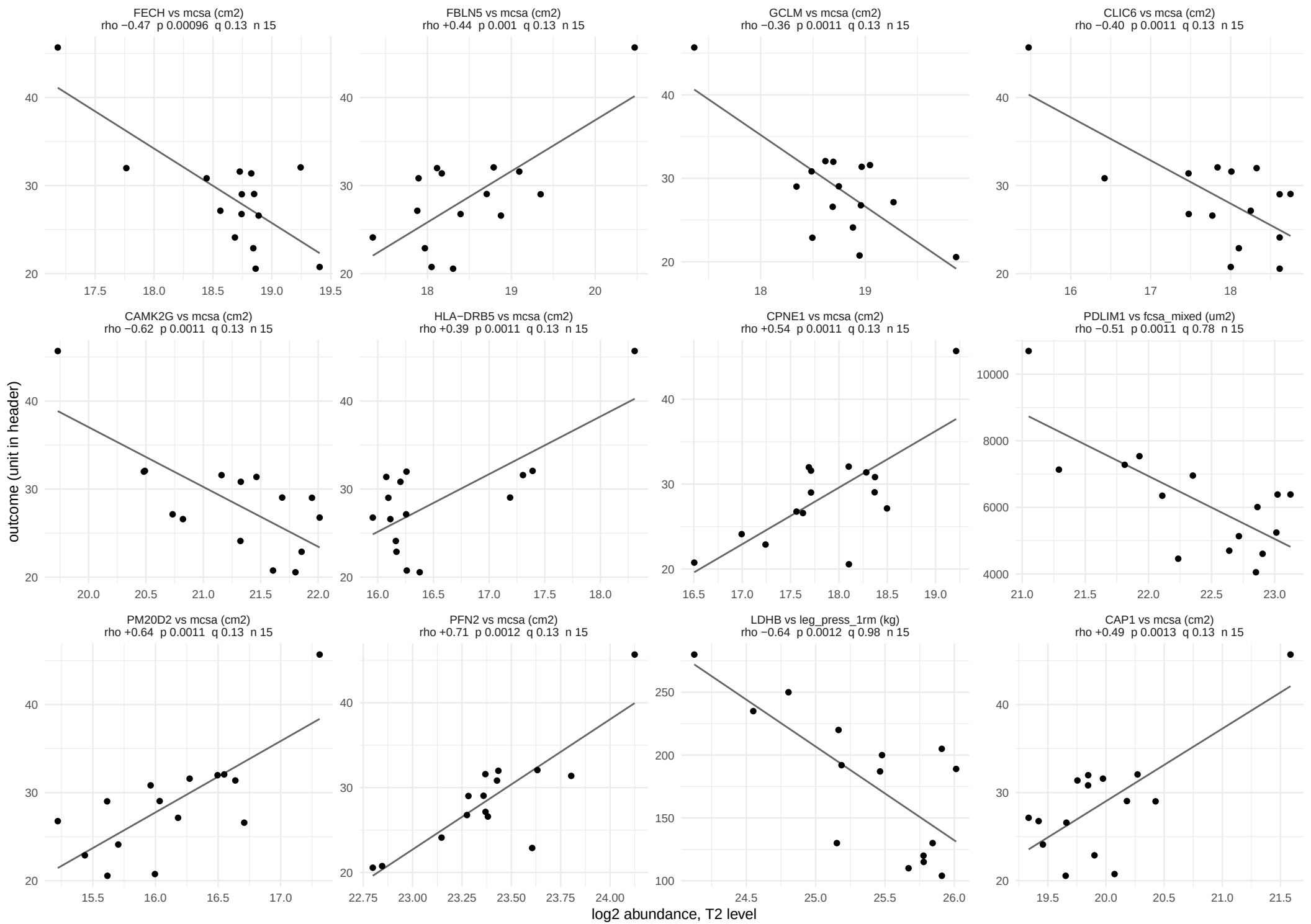
555 of 11400 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, post_level window

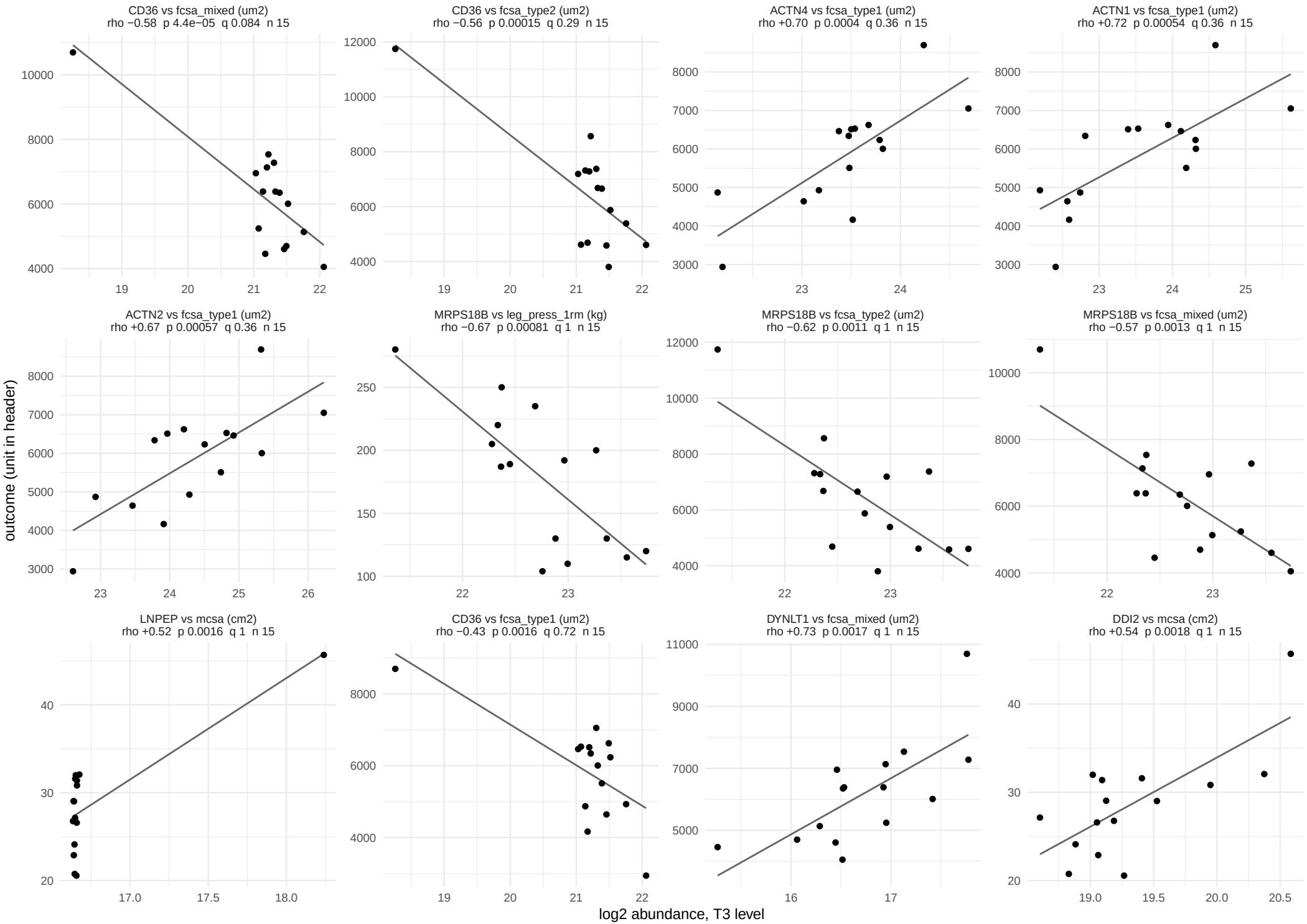
555 of 11400 feature−outcome pairs at p < 0.05, 1 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, post_bout_level window

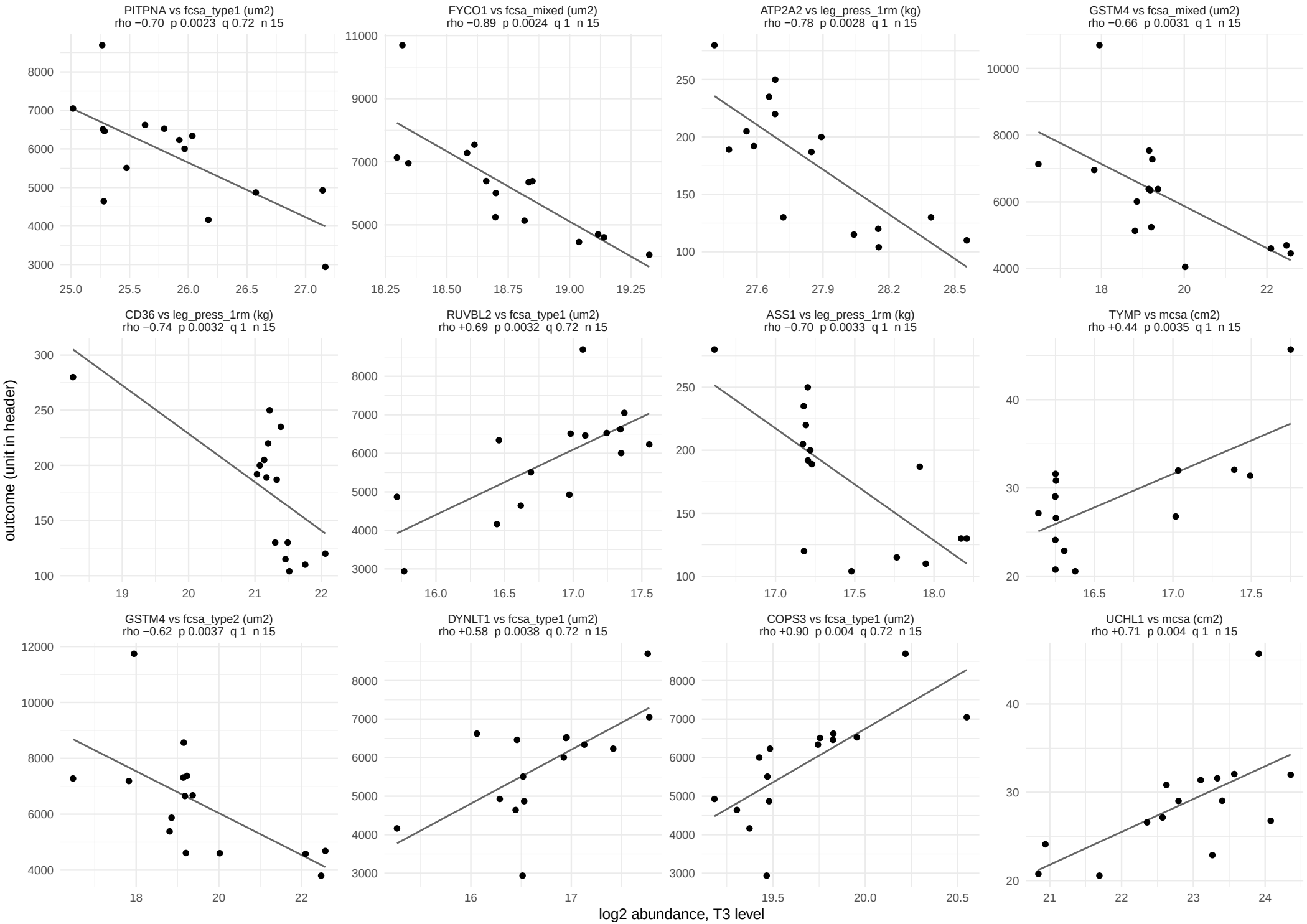
372 of 11400 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Change score, post_bout_level window

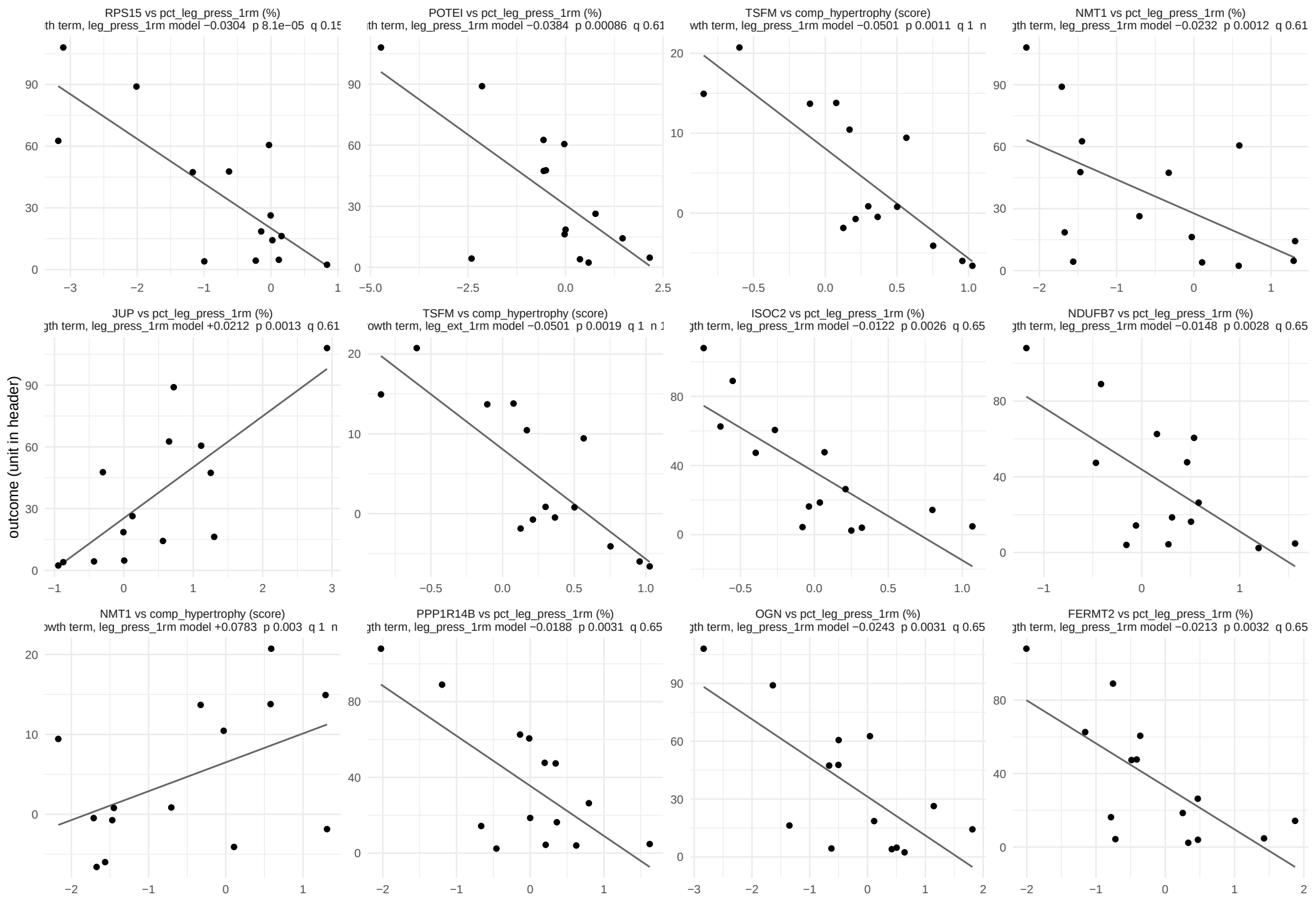
372 of 11400 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



One point per subject, every subject pooled. The line is a least−squares fit, drawn as a guide. Header: Spearman rho, and limma p and BH q within window, outcome, with the subjects used. Table: c_data/01_proteins.xlsx, change_score.

Two predictor, training window

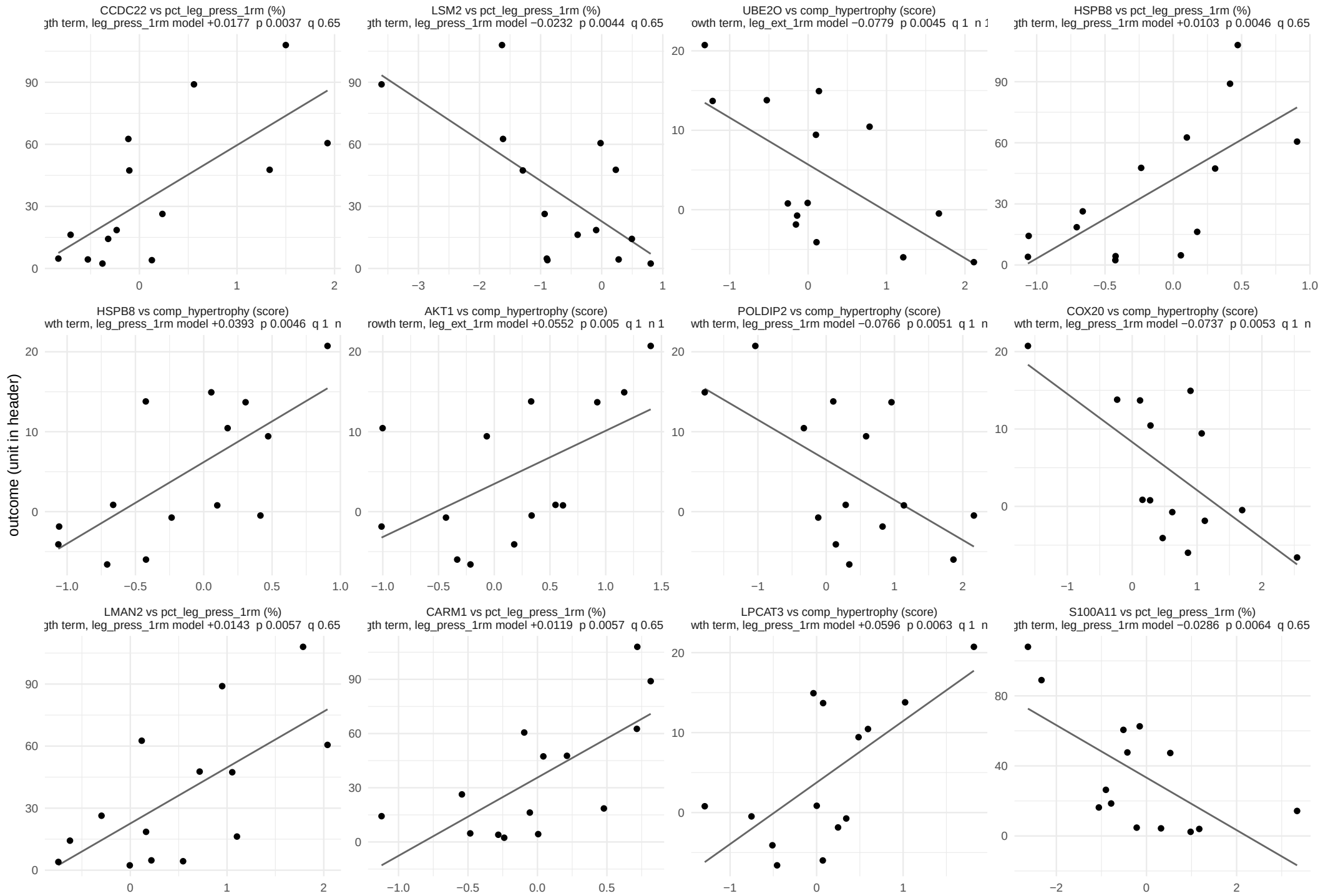
288 of 7600 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.

Two predictor, training window

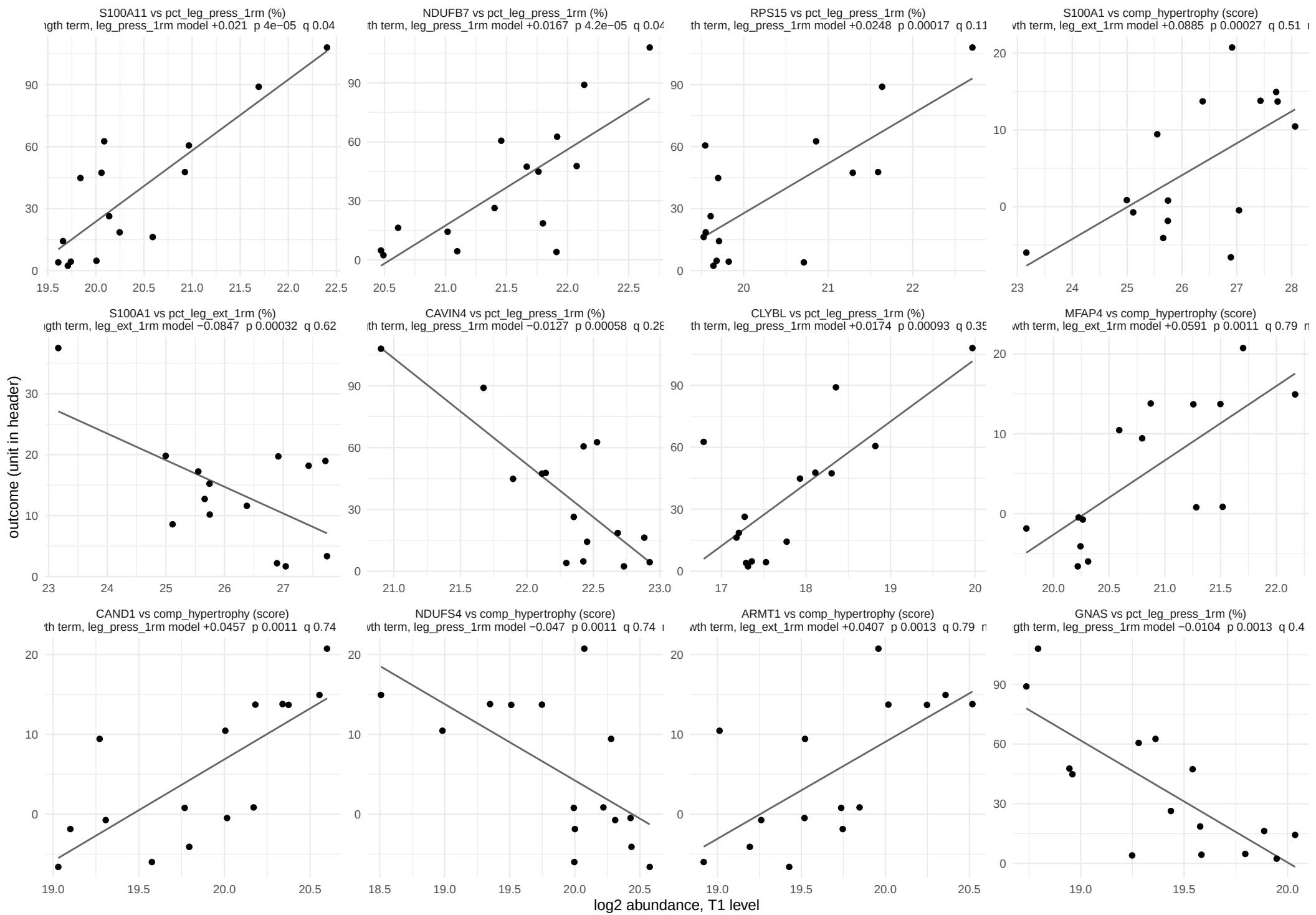
288 of 7600 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.

Two predictor, baseline window

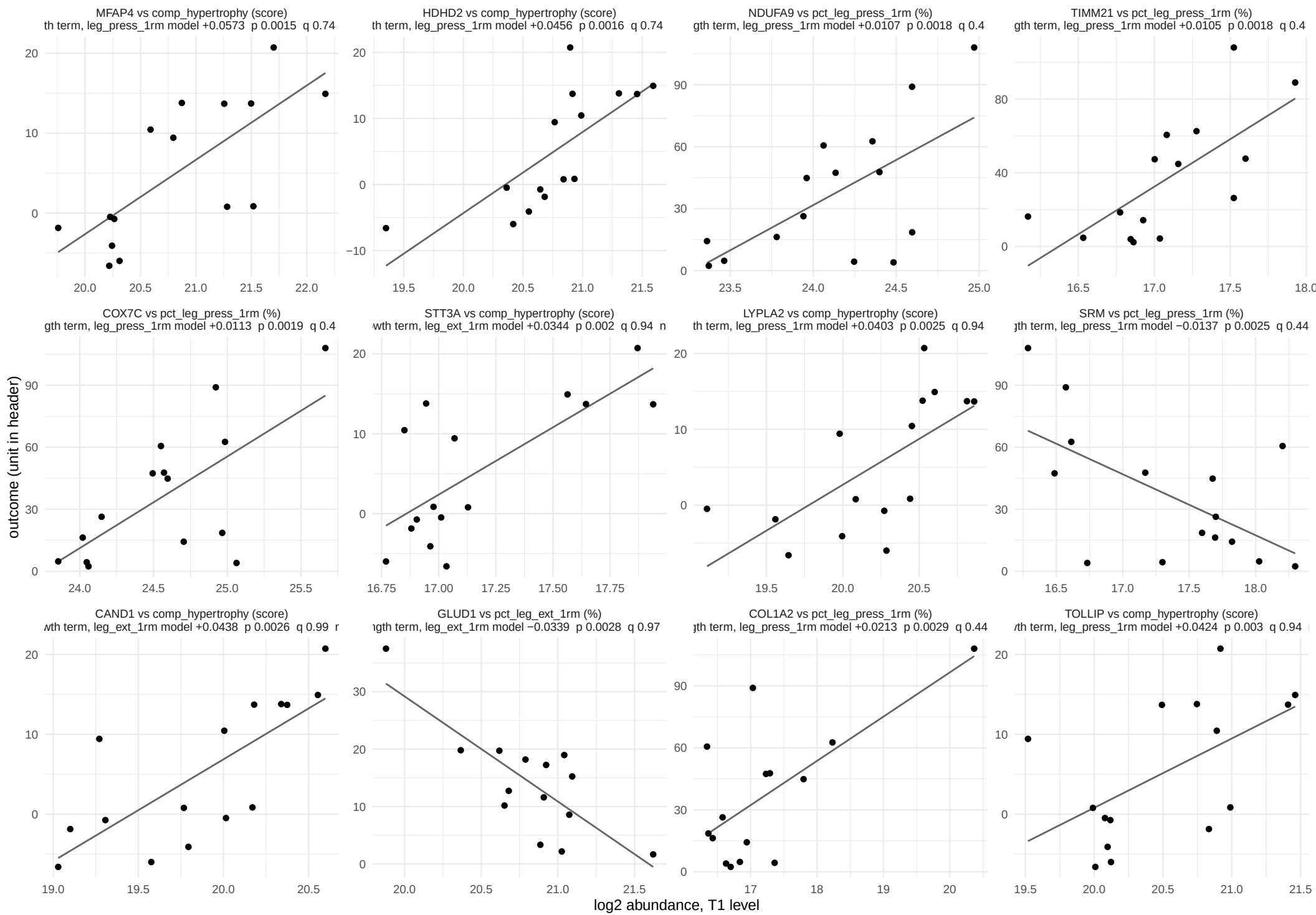
391 of 7600 feature−outcome pairs at p < 0.05, 2 at BH < 0.05. Page 1 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.

Two predictor, baseline window

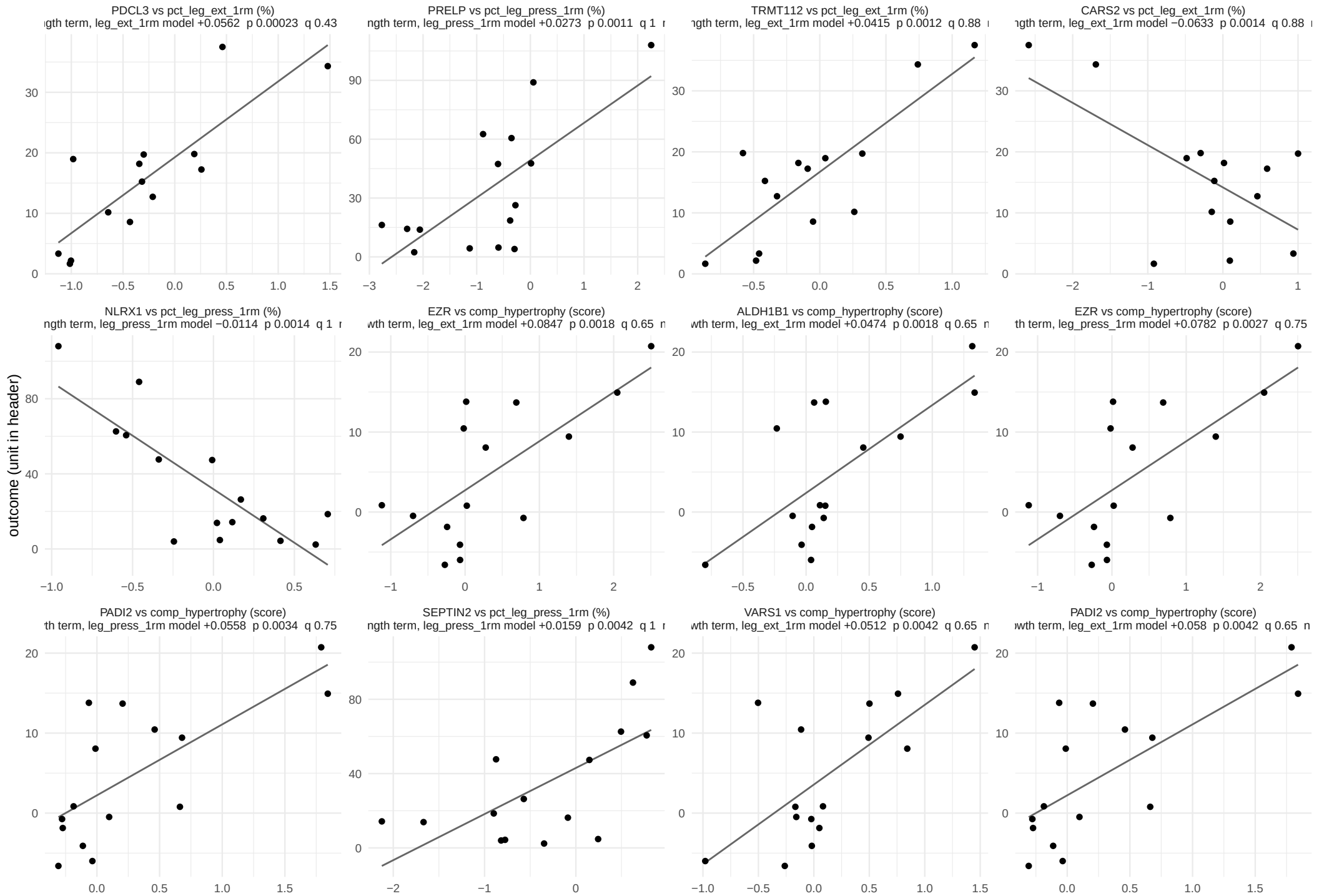
391 of 7600 feature−outcome pairs at p < 0.05, 2 at BH < 0.05. Page 2 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.

Two predictor, acute window

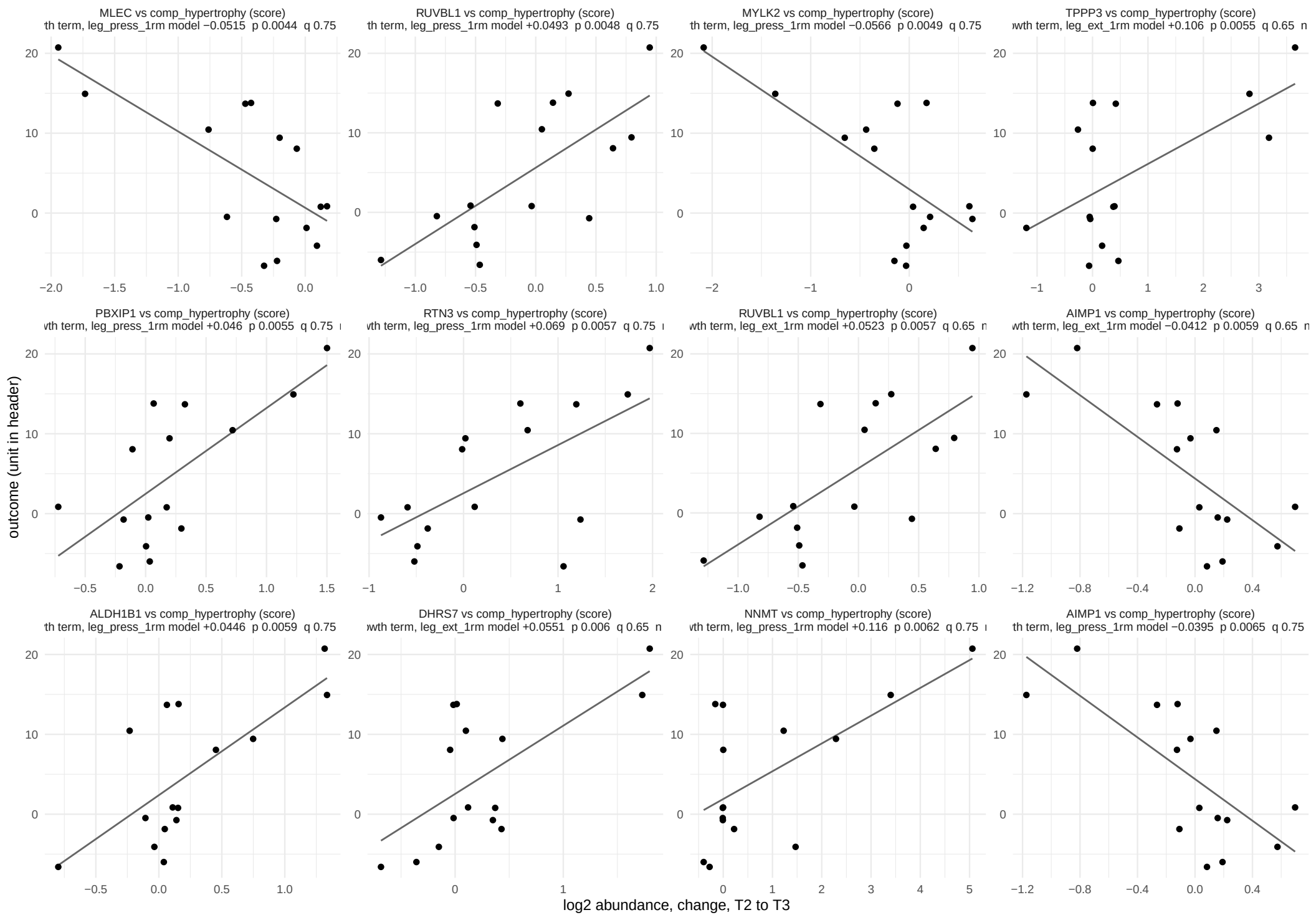
324 of 7600 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 1 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.

Two predictor, acute window

324 of 7600 feature−outcome pairs at p < 0.05, 0 at BH < 0.05. Page 2 of 2.



One point per subject. Header: the growth or strength coefficient from limma, feature ~ comp_hypertrophy + strength gain, each adjusted for the other, with p, BH q within window, strength test, term, and the subjects used. The x axis shows the term's own outcome; the drawing does not hold the other predictor fixed. Table: c_data/01_proteins.xlsx, two_predictor.